COMPLEXITY OF DEEP COMPUTATIONS VIA TOPOLOGY OF FUNCTION SPACES

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Abstract. We bring the topology of spaces of functions of the first Baire class, culminating in Todorčević's classification of Rosenthal compacta, together with Shelah's model-theoretic notion of independence, to bear on the complexity of deep computations, that is, of limits of computations. Such limits arise whenever an iterative procedure is carried to its conclusion, from the classical methods of numerical analysis, such as Newton's method and the power method, to the recurrent, equilibrium and continuous-depth architectures of deep learning; yet a deep computation is, in general, a nonconstructive object, an ultrafilter limit of possibly uncountably many computations that need not be the limit of any sequence of them. We show that, when states are observed through countably many real-valued features, a set of computations lacks Shelah's *independence property*, a dividing line of model theory akin to the Vapnik–Chervonenkis dimension of statistical learning, exactly when its deep computations are of the first Baire class on every region of the state space on which the features are bounded; on each such region, every deep computation is then the limit of a sequence of computations. Todorčević's classification then yields a strict hierarchy of complexity classes of deep computations. Its levels are attained, from the top down, by a rotation of the circle, the power method, Bresenham's line algorithm in the plane and in space, and the Euclidean algorithm, while Newton's method for the square roots of unity on the Riemann sphere lies beyond the hierarchy. For algorithms that iterate continuous steps on a compact metrizable space of states, the hierarchy coincides with the hierarchy of tame dynamical systems of Glasner and Megrelishvili. Finally, for any number of features, the absence of the independence property ensures that every deep computation can be evaluated almost surely by random sampling.

Introduction

This paper brings Shelah's theory of independence and Todorčević's classification of Rosenthal compacta to bear on the complexity of deep computations, the limits of computations: with these tools we determine when a deep computation is the limit of a sequence of computations, and we stratify deep computations into a strict hierarchy of complexity classes. The interest of the undertaking lies in the

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circumstance that a deep computation need not inherit the properties of the computations that converge to it: a pointwise limit of continuous maps need not be continuous, and a limit taken along an ultrafilter over uncountably many computations need not be the limit of any sequence of them, in which case it lies beyond the reach of sequential approximation. Deep computations arise whenever an iterative procedure is carried to its conclusion. The classical methods of numerical analysis, such as Newton's method for the roots of a polynomial and the power method for the eigenvectors of a matrix, furnish instances whose deep computations we analyze in Sections 4 and 5. In deep learning, several architectures are defined as limits of computations. A recurrent network applies one layer over and over, and the deep equilibrium models of Bai, Kolter and Koltun [BKK] compute directly, by root finding, the fixed point that an infinitely deep network of this kind, with the same weights in every layer, would reach. A residual network may be regarded as a discretization of a differential equation, and the neural ordinary differential equations of Chen, Rubanova, Bettencourt and Duvenaud [CRBD] replace its discrete sequence of layers by a differential equation for the hidden state, obtained in the limit in which the time step between consecutive layers tends to zero. Neural ordinary differential equations belong to a broader effort to bring differential equations into deep learning, which also includes the physics-informed neural networks of Raissi, Perdikaris and Karniadakis [RPK19], in which the differential equations of a physical system are built into the training of a network. In these models, as in the classical methods, the object of interest is a limit of computations rather than a single computation.

The framework of this paper is that of compositional computation structures, introduced by Alva, Dueñez, Iovino and Walton in the first paper of this series [ADIW26] to accommodate all these limits in a single setting. The framework concerns classical models of computation, in which real-valued quantities are known and manipulated to a finite degree of precision; computations act on states, and each state is observed through the values of real-valued *features*. In that paper the authors introduced two closely related notions of limit of computations: *ultracomputations*, so called because they arise as ultrafilter limits of standard computations, and *deep computations*, so called after the usage in machine learning [BKK]. The two notions differ in a technical point that plays no role here, and we speak of deep computations throughout. A set of computations is *stable* when all its deep computations are continuous on every *shard*, that is, on every set of states on which each feature obeys a prescribed bound. The analysis of stable sets rests on a theorem proved by Grothendieck in 1952 [Gro52], his double limit criterion: a uniformly bounded set of continuous functions on a compact space has only continuous functions in its pointwise closure if and only if, for every sequence of functions from the set and every sequence of points of the space, the two iterated limits agree whenever both exist. Using this criterion, Alva, Dueñez, Iovino and Walton characterized the stable sets by an exchange of iterated limits and showed that their deep computations are *polynomially approximable*: on every shard, each feature of their output is a uniform limit of polynomials in the features of the input. Polynomial approximability corresponds to continuous extendibility in topology, and to *stability* and *type definability* in model theory.

Beyond stability, Shelah's classification theory [She90] singles out the *independence property*, which Shelah isolated in 1971 [She71, Definition 4.1]. A formula

has the independence property when, for every natural number n, there are n parameters on which every pattern of truth and falsity of the formula is realized by some element of a model; a theory none of whose formulas has the independence property is called *NIP*, or *dependent*. The same dividing line appeared independently, within a few years, in two other fields, as a dichotomy between polynomial and exponential growth: a family of sets either shatters arbitrarily large finite sets, in the sense that it cuts out all their subsets, or leaves on every finite set a number of traces (intersections with members of the family) bounded by a polynomial in the size of the set. In statistical learning, Vapnik and Chervonenkis announced this dichotomy in 1968 and published its proof in 1971 [VC71], in their study of the uniform convergence of relative frequencies to probabilities; in combinatorics, Sauer [Sau72] and Shelah [She72] proved it in 1972, and it is now known as the Sauer–Shelah lemma. The largest size of a finite set shattered by a family is called the *Vapnik–Chervonenkis dimension* of the family.

In analysis, the same combinatorial pattern emerged from the study of Banach spaces. Rosenthal's ℓ_1-theorem of 1974 [Ros74] states that every bounded sequence in a Banach space has a subsequence that is either weakly Cauchy or equivalent to the unit vector basis of ℓ_1, the space of absolutely summable sequences. At the core of the proof of the ℓ_1-theorem is a dichotomy for bounded sequences of real functions on a set: such a sequence either has a pointwise convergent subsequence or has an *independent* subsequence, one for which there are two thresholds such that, for any two disjoint finite sets of indices, at some point the functions indexed by the first set take values below the lower threshold and those indexed by the second set take values above the upper one. Bourgain, Fremlin and Talagrand [BFT78] passed in 1978 from sequences to sets, and from convergence to the first Baire class, which consists of the pointwise limits of sequences of continuous functions. They characterized, by a condition of the same kind, the pointwise bounded sets of continuous functions on a Polish space, that is, on a separable and completely metrizable space, whose pointwise closures consist of functions of the first Baire class (see Theorem 1.4); on a compact metrizable space, their condition is exactly the absence of the independence property for the set of functions (see Theorem 1.6). Compact sets of functions of the first Baire class on Polish spaces, with the topology of pointwise convergence, were named *Rosenthal compacta* by Godefroy [God80], and they have since been studied intensively in functional analysis, especially in the theory of Banach spaces.

The classification of Rosenthal compacta took a decisive form in 1999, when Todorčević [Tod99] proved his trichotomy: every separable Rosenthal compactum that is not metrizable contains a copy of one of a few canonical compacta, and which of them occur depends on whether the compactum is hereditarily separable, first countable but not hereditarily separable, or not first countable (see Theorem 3.3). Todorčević's proof relies on set-theoretic forcing and infinite Ramsey theory, and his methods continue to be analyzed and exploited [Tod23, HT19]. Glasner and Megrelishvili [GM22] used the trichotomy to arrange into a hierarchy the tame dynamical systems on compact metrizable spaces [Köh95, Gla06], those whose enveloping semigroup, the pointwise closure of the set of maps of the system, consists of maps of the first Baire class; the levels of their hierarchy are defined by first countability, hereditary separability and metrizability of the enveloping semigroup.

The theories just recalled are the instruments with which we classify the unstable sets of computations. An unstable set of computations is *wild* when it has Shelah's *independence property* (IP), which means, roughly, that for some feature the computations realize every finite pattern of high and low values on suitable inputs, and it is *tame*, or *NIP*, when it does not have the independence property. Between stability and NIP we identify three new levels of tameness, NIP_3, NIP_2 and NIP_1, which together with NIP form a hierarchy of four tame classes beyond the stable ones. In this classification the theory of Rosenthal compacta assumes, on the topological side, the role that Grothendieck's criterion plays for stable sets, while on the model-theoretic side the role of stability passes to NIP, a central notion of the part of classification theory beyond stability that is known as *neostability*.

Model theory is the medium through which topology and computation are translated into each other. With every intermediate state of a computation we associate its *type*, the record of the real values that all the features of the computation take at that state. The types, together with their limits, form a space of real-valued functions on the set of features, the *space of types*, on which computations act; the embedding of a structure into its space of types is a central device of model theory, and it allows us to study topological limits of computations by means of the theory of function spaces with the topology of pointwise convergence, known as $\mathrm{C_p}$*-theory*. The space of types thus becomes a "Rosetta stone" connecting three classification programs: in topology, the classification of Rosenthal compacta pioneered by Todorčević [Tod99]; in logic, the classification of theories developed by Shelah [She71, She90]; and in statistical learning, the theory of Vapnik–Chervonenkis (VC) dimension and of PAC (probably approximately correct) learnability [VC71, VC74, Val84]. In Theorem 3.1 we show that, under this translation, NIP sets of computations correspond to spaces of deep computations that are Rosenthal compacta on each shard. On the side of learning, finiteness of VC dimension implies NIP, and the two notions coincide for first-order formulas (see Section 1.5).

Argyros, Dodos and Kanellopoulos [ADK08] refined Todorčević's trichotomy into a heptachotomy, which we likewise bring to bear on deep computations. It concerns families of functions indexed by the binary tree, compared along dyadic subtrees, the copies of the binary tree inside itself; two such families are *equivalent* when the bijection between them that respects the indexing extends to a homeomorphism of their closures. Argyros, Dodos and Kanellopoulos exhibited seven pairwise nonequivalent families of Baire class 1 functions indexed by the binary tree, each of them *minimal* in the sense that it is equivalent to each of its subfamilies indexed by a dyadic subtree. They proved that every relatively compact family of Baire class 1 functions on a Polish space, indexed by the binary tree, contains, along a dyadic subtree, a family equivalent to one of the seven.

Approximation by sequences rests on a property shared by two settings: that of Grothendieck [Gro52], in which the functions are continuous on a compact space, and that of Bourgain, Fremlin and Talagrand [BFT78], in which they are of Baire class 1 on a Polish space. In both settings, *compactness and sequential compactness coincide* for sets of such functions with the topology of pointwise convergence, and if the closure of a set of such functions is compact and consists of such functions, then every point of the closure is the limit of a sequence from the set. Consequently, when the features are countably many and the set of computations is NIP, every

deep computation is, on each shard, the limit of a *sequence* of standard computations, although in principle it is a nonconstructive ultrafilter limit over a possibly uncountable set of computations.

Our results show how the classification invariants used in computation, topology, descriptive set theory and model theory translate into one another. The paper presupposes no acquaintance with advanced topology, model theory or the theory of computation; undergraduate topology and measure theory suffice, and the further background is assembled in Section 1.

Our first main result, Theorem 3.1, concerns compositional computation structures with countably many features. In it we show that a set of computations has the NIP on every shard, one feature at a time, exactly when its deep computations form a Rosenthal compactum on each shard, that is, are of the first Baire class there. Guided by Todorčević's trichotomy, we then define three levels of tameness by the topology of these compacta (see Definition 3.2): a countable NIP set of computations is NIP_3, NIP_2 or NIP_1 when, on each shard, its deep computations form a metrizable, a hereditarily separable or a first countable space, respectively. The levels form the hierarchy

$$\text{stable} \Rightarrow \mathrm{NIP}_3 \Rightarrow \mathrm{NIP}_2 \Rightarrow \mathrm{NIP}_1 \Rightarrow \mathrm{NIP},$$

and in Section 5 we show that every implication in it is strict.

In that section we realize each level by a classical algorithm and exhibit an algorithm beyond the hierarchy:

- a rotation of the circle is stable;
- the power method is NIP_3 but not stable;
- Bresenham's line algorithm is
  - NIP_2 but not NIP_3 for lines in the plane,
  - NIP_1 but not NIP_2 for lines in three-dimensional space;
- the Euclidean algorithm, acting on the projective line by the group that its steps generate, is NIP but not NIP_1, and its deep computations include exact equality tests;
- Newton's method for the roots of $z^2 - 1$ on the Riemann sphere is not NIP.

For iterative algorithms, the hierarchy coincides with the hierarchy of tame dynamical systems studied by Glasner and Megrelishvili [GM22]. At the level NIP but not NIP_1, our two examples, the Euclidean algorithm and free reduction in the free group on two generators, are actions of groups with two generators rather than iterations of a single map; whether a single iterated map can reach this level is an open question, closely related to a question of Glasner and Megrelishvili (see Question 7.2).

The heptachotomy of Argyros, Dodos and Kanellopoulos refines this picture for families of computations indexed by the binary tree: on each shard and for each feature, every tame family of this kind contains, along a dyadic subtree, a family equivalent to one of the seven minimal families, which we read as primitives of binary search (see Section 3.3). Finally, in Theorem 6.7 and Corollary 6.14 we relate NIP and *Talagrand stability*, a measure-theoretic condition introduced by Talagrand in his study of Pettis integration, to the Monte Carlo computability of deep computations, that is, to their evaluation almost surely by random sampling.

Organization of the paper. In Section 1 we present the topological and combinatorial preliminaries, including five canonical Rosenthal compacta (Section 1.4) and

the role of NIP in model theory, combinatorics and learning (Section 1.5). In Section 2 we recall compositional computation structures, the source of the structural and model-theoretic viewpoint of the paper, and their stable sets of computations; the section ends with two motivating examples, Newton's method for square roots and Newton's method on the Riemann sphere, whose different behaviors the classification is designed to separate (Section 2.5). Section 3 contains the classification of deep computations and the main theorems. In Section 4 we work out Newton's method on the Riemann sphere and on the real projective line, and in Section 5 we place classical algorithms at every level of the hierarchy and beyond it. In Section 6 we replace topology by measure and study deep computations that can be evaluated almost surely by random sampling; we intend to develop this measure-theoretic viewpoint further in future research. Section 7 contains the questions that the paper leaves open.

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1. Topological preliminaries: From continuity to Baire class 1

In this section we assemble the topology of function spaces on which the rest of the paper rests. Deep computations are pointwise limits of continuous maps, and the two themes of this paper, the approximation of a deep computation by a sequence of computations and the complexity of the set of all deep computations, therefore belong to the theory of spaces of functions with the topology of pointwise convergence. We recall, in turn, Polish spaces and Baire category, the first Baire class, the dichotomies of Rosenthal and of Bourgain, Fremlin and Talagrand from which the NIP emerges, and the Rosenthal compacta, five of which serve as yardsticks for the classification; the section closes with the role of the NIP in model theory, combinatorics and statistical learning. All spaces are assumed to be Hausdorff unless explicitly stated otherwise.

We write $\mathbb{N}$ for the set $\{0, 1, 2, \dots\}$ of natural numbers, so that the semigroups of iterates considered below contain the identity. A subset of a topological space is F_σ if it is a countable union of closed sets, and G_δ if it is a countable intersection of open sets. In a metric space every open set is F_σ, and, equivalently, every closed set is G_δ.

A *Polish space* is a separable topological space whose topology is induced by a complete metric. Countable products of Polish spaces are Polish. Classical examples, besides the real line, are the Cantor space $2^{\mathbb{N}}$ of all infinite binary sequences, the Baire space $\mathbb{N}^{\mathbb{N}}$ of all infinite sequences of natural numbers, and the space $\mathbb{R}^{\mathbb{N}}$ of all sequences of real numbers, each with the product topology.

Fact 1.1. [Eng89, 4.3.24] *A subset of a Polish space is itself Polish in the subspace topology if and only if it is a G_δ set. In particular, closed subsets and open subsets of Polish spaces are also Polish spaces.*

A subset A of a topological space X is *nowhere dense* if its closure has empty interior, and of the *first category*, or *meager*, if it is contained in a countable union of closed nowhere dense sets; otherwise A is of the *second category*. The complements of meager sets are called *comeager*. Since the closure of a nowhere dense set is nowhere dense, the meager subsets of X form the σ-ideal generated by the nowhere dense subsets of X. A topological space X is *Baire* if every comeager subset of X is dense, or, equivalently, if every countable intersection of dense open sets is dense.

The Baire category theorem states that compact Hausdorff spaces and completely metrizable spaces, Polish spaces in particular, are Baire.[1]

For topological spaces X and Y, we denote by $\mathrm{C_p}(X,Y)$ the set of all continuous functions $f\colon X \to Y$ with the topology of pointwise convergence, that is, the topology it inherits as a subspace of the product Y^X, in which X serves merely as an index set. We write $\mathrm{C_p}(X)$ for $\mathrm{C_p}(X,\mathbb{R})$.

A function $f\colon X \to Y$ between topological spaces is of *Baire class 1* if it is the pointwise limit of a sequence $(f_n)_{n\in\mathbb{N}}$ of continuous functions, that is, $f(x) = \lim_{n\to\infty} f_n(x)$ for all $x \in X$. When Y is metrizable, an elementary argument shows that a function of Baire class 1 is continuous at every point outside a set of the first category. We denote by $\mathrm{B_1}(X,Y)$ the set of functions $X \to Y$ of Baire class 1 with the topology of pointwise convergence, and we write $\mathrm{B_1}(X)$ for $\mathrm{B_1}(X,\mathbb{R})$; the inclusions $\mathrm{C_p}(X,Y) \subseteq \mathrm{B_1}(X,Y) \subseteq Y^X$ are topological. Relatively compact subsets of $\mathrm{C_p}(X)$ and of $\mathrm{B_1}(X)$, especially for Polish X, are central objects of study in functional analysis and in topological dynamics.

A topological space X is *perfectly normal* if it is normal and every closed subset of X is a G_δ set (equivalently, every open subset of X is an F_σ set). Every metrizable space (hence, every Polish space) is perfectly normal.

The following characterization of Baire class 1 functions is classical; for a proof, see [Tod97, Section 10].

Fact 1.2 (Lebesgue, Hausdorff, Baire[2])**.** *If X is perfectly normal, then the following conditions are equivalent for a function $f : X \to \mathbb{R}$:*

(1) *f is a Baire class 1 function, that is, f is a pointwise limit of continuous functions.*
(2) *$f^{-1}[U]$ is an F_σ subset of X whenever $U \subseteq \mathbb{R}$ is open.*

If X is Polish, then (1) and (2) are equivalent to:

(3) *For every nonempty closed $F \subseteq X$, the restriction $f|_F$ has a point of continuity.*

Moreover, if X is Polish and $f \notin \mathrm{B_1}(X)$, then there exist countable $D_0, D_1 \subseteq X$ and reals $a < b$ such that

$$D_0 \subseteq f^{-1}(-\infty, a], \quad D_1 \subseteq f^{-1}[b,\infty), \quad \overline{D_0} = \overline{D_1}.$$

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[1] More generally, any Čech-complete space is Baire [Eng89, Theorem 3.9.3].

[2] The implication (1)⇒(2) holds in every topological space. Indeed, if f is the pointwise limit of a sequence $(f_n)_{n\in\mathbb{N}}$ of continuous functions and a is a real number, then

$$f^{-1}(a,\infty) = \bigcup_{k\geq 1} \bigcup_{m\in\mathbb{N}} \bigcap_{n\geq m} f_n^{-1}[a + 1/k, \infty)$$

is a countable union of closed sets; the same is true of $f^{-1}(-\infty, a)$, and condition (2) follows because every open subset of $\mathbb{R}$ is a countable union of intervals $(a,b) = (a,\infty) \cap (-\infty, b)$. The reverse implication, (2)⇒(1), was established by Lebesgue for real-valued functions of finitely many real variables [Leb04] and extended by Hausdorff to real-valued functions on metric spaces [Kec95, Theorem 24.10]. The argument requires of X only that it be normal and that each of its open subsets be an F_σ set, which are precisely the defining properties of a perfectly normal space. The implication (1)⇒(3) holds whenever X is Polish. If f satisfies (1) and F is a nonempty closed subset of X, the restriction $f|_F$ is of Baire class 1, and its points of discontinuity therefore form a meager subset of F, as noted before Fact 1.2. Since F, being closed in a Polish space, is itself Polish and hence a Baire space, this meager set cannot exhaust F, and so $f|_F$ has a point of continuity. The reverse implication, (3)⇒(1), is due to Baire, who proved it in his thesis for real-valued functions of real variables [Bai99]; for Polish spaces, see [Kec95, Theorem 24.15].

1.1. **From Rosenthal's dichotomy to the Bourgain–Fremlin–Talagrand dichotomy to Shelah's NIP.** The question when a deep computation is accessible to approximation, that is, when it is the limit of a *sequence* of standard computations, is answered by the dichotomies of Rosenthal and of Bourgain, Fremlin and Talagrand, which we recall in this subsection and from which the NIP emerges in the theory of function spaces. In a metrizable space every point of the closure of a set is the limit of a sequence of points of the set; in spaces of functions with the topology of pointwise convergence the closure need not admit this sequential description, and the two dichotomies determine when it does not.

On the Cantor space $X = 2^{\mathbb{N}}$, the n-th coordinate map $p_n \colon X \to \{0,1\}$ is continuous for each $n \in \mathbb{N}$; however, the sequence $(p_n)_{n\in\mathbb{N}}$ has *no* convergent subsequence in $\mathbb{R}^X$ (see [Tod97, Chapter 1.1]). In a sense, this sequence exhibits the most extreme failure of sequential convergence: the closure of $\{p_n : n \in \mathbb{N}\}$ in $\{0,1\}^X$, or in $\mathbb{R}^X$, is homeomorphic to the Stone–Čech compactification $\beta\mathbb{N}$ of the discrete space of natural numbers. The space $\beta\mathbb{N}$ has cardinality $2^{2^{\aleph_0}}$, the largest possible for a separable Hausdorff space, and its points correspond to the ultrafilters on $\mathbb{N}$.[3]

The following theorem, proved by Haskell Rosenthal in 1974, shows that, for pointwise bounded sequences of continuous functions on a Polish space, a copy of $\beta\mathbb{N}$, such as the closure of the coordinate maps, is the only obstruction to the existence of a convergent subsequence.

Theorem 1.3 (Rosenthal's dichotomy [Ros74])**.** *If X is a Polish space and $(f_n)_{n\in\mathbb{N}}$ is a pointwise bounded sequence in $\mathrm{C_p}(X)$, then the sequence (f_n) has either a convergent subsequence or a subsequence whose closure in $\mathbb{R}^X$ is homeomorphic to $\beta\mathbb{N}$.*

The dichotomy grew out of *Rosenthal's ℓ_1-theorem*, which states that every bounded sequence in a Banach space has a subsequence that is either weakly Cauchy or equivalent to the unit vector basis of ℓ_1, the space of absolutely summable sequences; in particular, a Banach space contains no isomorphic copy of ℓ_1 if and only if every bounded sequence in it has a weakly Cauchy subsequence.

The theorem of Bourgain, Fremlin and Talagrand concerns sets rather than sequences: for a pointwise bounded set of continuous functions on a Polish space X, it characterizes the relative compactness of the set in the larger space $\mathrm{B}_1(X)$, that is, the property that every function in its pointwise closure be of Baire class 1. The theorem is a counterpart of Rosenthal's dichotomy, although it is usually stated as an equivalence rather than as a dichotomy (see Theorem 1.4 below).

The statement of the theorem requires the following notation. For $f \in \mathbb{R}^X$ and a real number a, define

$$X^f_{\leq a} \coloneqq f^{-1}(-\infty, a], \qquad\qquad X^f_{\geq a} \coloneqq f^{-1}[a, +\infty).$$

Finally, the last assertion of Fact 1.2 follows from the implication (3)⇒(1) together with the Baire Category Theorem.

[3]Addition on $\mathbb{N}$ extends to an associative operation on $\beta\mathbb{N}$, and, by the Ellis–Numakura lemma, $\beta\mathbb{N}$ has idempotent elements for this operation, the idempotent ultrafilters. Alva, Dueñez, Iovino and Walton [ADIW26] use them to prove the existence of *deep equilibria* in compositional computation structures, that is, of limits of the iterates of a single computation, taken along an ultrafilter on $\mathbb{N}$, that are idempotent under composition.

For a set $A \subseteq \mathbb{R}^X$ and a real number a, define, by a slight abuse of notation,

$$X^A_{\leq a} \coloneqq \bigcap_{f\in A} X^f_{\leq a} = \{x \in X : f(x) \leq a \text{ for all } f \in A\},$$
$$X^A_{\geq a} \coloneqq \bigcap_{f\in A} X^f_{\geq a} = \{x \in X : f(x) \geq a \text{ for all } f \in A\};$$

in particular, $X^{\emptyset}_{\geq a} = X = X^{\emptyset}_{\leq a}$. For a sequence $(f_n)_{n\in\mathbb{N}}$ in $\mathbb{R}^X$ and a set $I \subseteq \mathbb{N}$, define $I^{\complement} \coloneqq \mathbb{N} \setminus I$ and $f_I \coloneqq \{f_i : i \in I\}$.

Theorem 1.4 (The Bourgain–Fremlin–Talagrand dichotomy [BFT78])**.** *Let X be a Polish space, and let $A \subseteq \mathrm{C_p}(X)$ be pointwise bounded. The following are equivalent:*

(i) A is relatively compact in $\mathrm{B}_1(X)$.
(ii) For every $(f_n)_{n\in\mathbb{N}} \subseteq A$ and every $a < b$ there is $I \subseteq \mathbb{N}$ such that

$$X^{f_I}_{\leq a} \cap X^{f_{I^{\complement}}}_{\geq b} = \emptyset.$$

(In this form, the Bourgain–Fremlin–Talagrand dichotomy is a special case of the equivalence (ii)⇔(v) of Corollary 4G of Bourgain, Fremlin and Talagrand [BFT78].)

In condition (ii) of Theorem 1.4, the set $X^{f_I}_{\leq a}$ is defined by the inequalities $f_i(x) \leq a$ with $i \in I$, and the set $X^{f_{I^{\complement}}}_{\geq b}$ by the inequalities $f_j(x) \geq b$ with $j \in I^{\complement}$. Since I or $I^{\complement}$ is infinite, at least one of the two sets is defined by countably infinitely many inequalities. A computation, however, can check only finitely many inequalities at a time, and the following definitions accordingly impose only finitely many.

Definition 1.5. A set $A \subseteq \mathbb{R}^X$ of functions has the *finitary No-Independence Property (NIP)* if, for every sequence $(f_n)_{n\in\mathbb{N}}$ in A and all reals $a < b$, there exist finite disjoint sets $E, F \subseteq \mathbb{N}$ such that $X^{f_E}_{\leq a} \cap X^{f_F}_{\geq b} = \emptyset$; we say that such sets E, F *witness finitary NIP for (f_n) and a, b.*

A set $A \subseteq \mathbb{R}^X$ has the *finitary Independence Property (IP)* if it does not have finitary NIP, that is, if there exist a sequence $(f_n)_{n\in\mathbb{N}}$ in A and reals $a < b$ such that $X^{f_E}_{\leq a} \cap X^{f_F}_{\geq b} \neq \emptyset$ for every pair of finite disjoint sets $E, F \subseteq \mathbb{N}$.

If the word "finite" is omitted in these definitions, we obtain *countable* NIP, which is weaker than finitary NIP, and *countable* IP, which is stronger than finitary IP. If, in addition, the witnesses are required to satisfy $F = E^{\complement}$, we obtain the properties that we call *BFT-NIP* and *BFT-IP*. Theorem 1.4 thus states that, on a Polish space X, a pointwise bounded set $A \subseteq \mathrm{C_p}(X)$ is relatively compact in $\mathrm{B}_1(X)$ if and only if it has BFT-NIP.

From now on, IP and NIP mean *finitary* IP and NIP unless otherwise specified.

Theorem 1.6. *Let X be a compact metrizable (hence Polish) space. For every pointwise bounded set $A \subseteq \mathrm{C_p}(X)$, the following are equivalent:*

(i) A is relatively compact in $\mathrm{B}_1(X)$;
(ii) A has BFT-NIP;
(iii) A has countable NIP;
(iv) A has finitary NIP.

(The equivalence *(ii)⇔(iv)* holds for every compact space X, even without the assumption that A is pointwise bounded, and the equivalence *(ii)⇔(iii)* holds for every set $A \subseteq \mathbb{R}^X$ on an arbitrary space X.)

Proof. The equivalence (i)⇔(ii) is Theorem 1.4. The implications (iv)⇒(iii) and (ii)⇒(iii) are trivial, and (iii)⇒(ii) follows by taking $I = E$, since then $F \subseteq I^{\complement}$. To prove (ii)⇒(iv), suppose that (f_n) and $a < b$ witness finitary IP, and let $I \subseteq \mathbb{N}$ be given. The closed sets $X^{f_n}_{\leq a}$ $(n \in I)$ and $X^{f_m}_{\geq b}$ $(m \in I^{\complement})$ have the finite intersection property, so, by compactness, $X^{f_I}_{\leq a} \cap X^{f_{I^{\complement}}}_{\geq b} \neq \emptyset$. Since I is arbitrary, A does not have BFT-NIP, and (ii) fails. □

Theorem 1.6 can be read as a dichotomy: if X is compact metrizable and $A \subseteq \mathrm{C_p}(X)$ is pointwise bounded, then either A is relatively compact in $\mathrm{B}_1(X)$, or A has IP, in the finitary and, equivalently, in the countable sense.

The independence property was isolated by Saharon Shelah [She71, Definition 4.1] as one of the dividing lines of his classification of first-order theories [She90]. We return to it in Section 1.5, where it appears together with its counterparts in combinatorics and in statistical learning.

1.2. **Rosenthal compacta.** The Bourgain–Fremlin–Talagrand dichotomy leads to the Rosenthal compacta, the compact spaces within which the deep computations of this paper are classified, and we recall them here. By Theorem 1.4, if a pointwise bounded set of continuous functions on a Polish space has BFT-NIP, then its pointwise closure is a compact set of functions of the first Baire class. Compact sets of this kind had appeared in Rosenthal's work on the ℓ_1-theorem, and Godefroy [God80] named them after him in the following definition.

Definition 1.7. A *Rosenthal compactum* is a topological space homeomorphic to a compact subset of $\mathrm{B}_1(X)$, with the topology of pointwise convergence, for some Polish space X.

Rosenthal compacta have strong tameness properties; for instance, every point in the closure of a subset of a Rosenthal compactum is the limit of a sequence of points of the subset [BFT78, Theorem 3F]. In this paper they serve as spaces of deep computations: in Theorem 3.1 below we show that, for countably many features, the deep computations of an NIP set of computations form a Rosenthal compactum on each shard.

1.3. **The special case $\mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}})$ with $\mathcal{P}$ countable.** This subsection is devoted to reducing the study of maps with values in a product of copies of the real line to that of real-valued functions. The reduction is needed because a state of a computation is described by the values of its features, and a computation is therefore a map with values in the product of copies of the real line indexed by the features. Fix a countable set $\mathcal{P}$ (in this paper, "countable" means finite or countably infinite), whose elements we call *predicate symbols* or *formal predicates*. Our goal is to characterize the relatively compact subsets of $\mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}})$, where X is a perfectly normal space, typically a Polish space.

Whenever $\mathcal{P}$ is regarded as a topological space, it carries the discrete topology, and the product $X \times \mathcal{P}$ is regarded either as a set or as a topological product, depending on the context. The spaces $\mathrm{C_p}(X, \mathbb{R}^{\mathcal{P}}) \subseteq \mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}})$ are subspaces of the ambient space $(\mathbb{R}^{\mathcal{P}})^X$, which can be described in two other ways, through the natural homeomorphisms

$$(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X \times \mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}},$$

given by

$$\mathbb{R}^{X\times\mathcal{P}} \to (\mathbb{R}^{\mathcal{P}})^X : \varphi \mapsto \hat{\varphi}$$
$$\mathbb{R}^{X\times\mathcal{P}} \to (\mathbb{R}^X)^{\mathcal{P}} : \varphi \mapsto \check{\varphi},$$

where

$$\hat{\varphi}(x) \coloneqq \varphi(x,\cdot) \in \mathbb{R}^{\mathcal{P}}, \qquad \check{\varphi}(P) \coloneqq \varphi(\cdot,P) \in \mathbb{R}^X.$$

These identifications treat X and $\mathcal{P}$ as mere sets; in particular, the topology of X plays no role in them.

For $x \in X$, define the "left projection" map

$$\lambda_x : \mathbb{R}^{X\times\mathcal{P}} \to \mathbb{R}^{\mathcal{P}} : \varphi \mapsto \lambda_x(\varphi) \coloneqq \hat{\varphi}(x);$$

for $P \in \mathcal{P}$, the "right projection" map

$$\rho_P : \mathbb{R}^{X\times\mathcal{P}} \to \mathbb{R}^X : \varphi \mapsto \rho_P(\varphi) \coloneqq \check{\varphi}(P).$$

For fixed $x \in X$ and $P \in \mathcal{P}$, we also have canonical projection maps

$$\lambda_x : \mathbb{R}^X \to \mathbb{R} : f \mapsto f(x), \qquad \rho_P : \mathbb{R}^{\mathcal{P}} \to \mathbb{R} : f \mapsto f(P).$$

When the context precludes ambiguity, we replace the specific symbols "λ" (for left) and "ρ" (for right) by the generic symbol "π"; thus π_x may stand for λ_x, and π_P for ρ_P.

The following proposition reduces the study of continuous functions, and of functions of Baire class 1, from X to $\mathbb{R}^{\mathcal{P}}$ to the case of real-valued functions.

Proposition 1.8. *The identification* $(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X\times\mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}}$ *induces identifications*

$$\mathrm{C_p}(X,\mathbb{R}^{\mathcal{P}}) \cong \mathrm{C_p}(X\times\mathcal{P}) \cong \mathrm{C_p}(X)^{\mathcal{P}}, \quad \mathrm{B}_1(X,\mathbb{R}^{\mathcal{P}}) \cong \mathrm{B}_1(X\times\mathcal{P}) \cong \mathrm{B}_1(X)^{\mathcal{P}}.$$

(The cardinality of $\mathcal{P}$ plays no role here.)

Proof. The identification of the $\mathrm{C_p}$-spaces follows from the definition of the product topology and the discreteness of $\mathcal{P}$: a continuous map $X \to \mathbb{R}^{\mathcal{P}}$ is precisely a $\mathcal{P}$-indexed family of continuous functions $X \to \mathbb{R}$, and such families correspond to continuous functions $X \times \mathcal{P} \to \mathbb{R}$. The identification of the spaces of functions of Baire class 1 follows, since membership in the first Baire class is defined through pointwise limits of sequences of continuous functions, and a sequence of $\mathcal{P}$-indexed families of continuous functions is the same as a $\mathcal{P}$-indexed family of sequences of continuous functions. □

For $A \subseteq Y^X$ and $K \subseteq X$, we write $A|_K \coloneqq \{f|_K : f \in A\}$ for the set of restrictions to K of the functions in A; for $A \subseteq (\mathbb{R}^{\mathcal{P}})^X$ and $P \in \mathcal{P}$, we write $\pi_P \circ A \coloneqq \{\pi_P \circ f : f \in A\}$, where $\pi_P = \rho_P \colon \mathbb{R}^{\mathcal{P}} \to \mathbb{R}$ is the projection onto the coordinate P. The following theorem extends Theorem 1.6 to $\mathbb{R}^{\mathcal{P}}$-valued functions.

Theorem 1.9. *Assume that* $\mathcal{P}$ *is countable,* X *is a Polish space, and* $A \subseteq \mathrm{C_p}(X,\mathbb{R}^{\mathcal{P}})$ *is pointwise bounded in the sense that the set* $\pi_P \circ A \subseteq \mathrm{C_p}(X)$ *is pointwise bounded for every* $P \in \mathcal{P}$*. The following are equivalent for every compact* $K \subseteq X$*:*

(i) $A|_K$ *is relatively compact in* $\mathrm{B}_1(K,\mathbb{R}^{\mathcal{P}})$*;*
(ii) $\pi_P \circ A|_K$ *has NIP for every* $P \in \mathcal{P}$*.*

Proof. By Proposition 1.8 and Tychonov's theorem, condition (i) holds if and only if $\pi_P \circ A|_K$ is relatively compact in $\mathrm{B}_1(K)$ for every $P \in \mathcal{P}$. Since K is compact metrizable, Theorem 1.6 shows that this is equivalent to (ii). □

In the applications, the NIP is verified on a set of states, whereas compactness is available only on its closure; the following lemma shows that the NIP passes from a set to its closure.

Lemma 1.10. *Assume that X is Hausdorff and that $A \subseteq \mathrm{C_p}(X)$. The following are equivalent for every $L \subseteq X$:*

(i) $A|_L$ satisfies the NIP;
(ii) $A|_{\overline{L}}$ satisfies the NIP.

Proof. The implication (ii)⇒(i) is trivial. Conversely, suppose that a sequence (f_n) in A and reals $a < b$ witness the IP on $\overline{L}$, and choose a' and b' with $a < a' < b' < b$. For finite disjoint $E, F \subseteq \mathbb{N}$, the open set $\bigcap_{n\in E} f_n^{-1}(-\infty, a') \cap \bigcap_{n\in F} f_n^{-1}(b', \infty)$ contains a point of $\overline{L}$, namely a point at which $f_n \le a$ for $n \in E$ and $f_n \ge b$ for $n \in F$; hence it meets L. Therefore the sequence (f_n) and the reals $a' < b'$ witness the IP on L. □

1.4. **Five canonical Rosenthal compacta.** Here we fix the notation for five Rosenthal compacta, all built on the Cantor set, that are the canonical members of the classes of Todorčević's trichotomy (see Theorem 3.3 below); by the trichotomy, the space of deep computations of each non-metrizable NIP example of Section 5 contains a copy of one of them. The five compacta are distinguished from one another by three topological properties. Recall that a topological space is *separable* if it has a countable dense subset, *hereditarily separable* if each of its subspaces is separable, and *first countable* if each of its points has a countable base of neighborhoods.

Let $2^{\mathbb{N}}$ carry the product topology and the lexicographic order $<$, whose order topology is the product topology. The set of finite binary words is denoted $2^{<\mathbb{N}}$, the length of a word $t \in 2^{<\mathbb{N}}$ is denoted $|t|$, and, for $a \in 2^{\mathbb{N}}$ and $n \in \mathbb{N}$, we write $a{\restriction}n$ for the word formed by the first n bits of a. Two words are *comparable* if one extends the other, and an *antichain* is a set of pairwise incomparable words. For a finite word $t \in 2^{<\mathbb{N}}$, let $t^\frown 0^\infty$ and $t^\frown 1^\infty$ be the sequences that begin with t and continue with all 0's and with all 1's, respectively; the set of sequences extending t is the clopen interval $[t^\frown 0^\infty, t^\frown 1^\infty]$.

For a statement S, we write $[\![S]\!]$ for its truth value: 1 if S is true and 0 if it is false. We introduce the following functions $2^{\mathbb{N}} \to \{0, 1\}$:

- the *prefix test* $v_t := [\![\cdot{\restriction}|t| = t]\!]$, the characteristic function of the set of sequences extending a fixed word $t \in 2^{<\mathbb{N}}$;
- the *equality test* $\delta_a := [\![\cdot = a]\!]$, the characteristic function of the singleton $\{a\}$ for fixed $a \in 2^{\mathbb{N}}$;
- the *threshold tests* $\chi_{[a} := [\![a \le \cdot]\!]$ and $\chi_{(a} := [\![a < \cdot]\!]$, the characteristic functions of the final intervals $\{x : a \le x\}$ (closed) and $\{x : a < x\}$ (open), for fixed $a \in 2^{\mathbb{N}}$; the special case $\mathbf{0} := \chi_{(1^\infty}$ is the identically zero function.

Prefix tests are continuous, and so are $\chi_{[t^\frown 0^\infty}$ and $\chi_{(t^\frown 1^\infty}$, since each of these functions depends only on the first $|t|$ bits of its argument; equality tests, and threshold tests at all other points, are discontinuous functions of Baire class 1. All closures below are taken in the topology of pointwise convergence.

(1) The *Alexandroff compactification* $A(2^{\mathbb{N}}) = \{\delta_a : a \in 2^{\mathbb{N}}\} \cup \{\mathbf{0}\}$. Each δ_a is isolated, and every neighborhood of $\mathbf{0}$ contains all but finitely many of the functions δ_a. Thus the Alexandroff compactification is neither separable nor first countable at $\mathbf{0}$.

(2) The *extended Alexandroff compactification* $\hat{A}(2^{\mathbb{N}}) = \overline{\{v_t : t \in 2^{<\mathbb{N}}\}} = \{v_t : t \in 2^{<\mathbb{N}}\} \cup A(2^{\mathbb{N}})$. As $n \to \infty$, the prefix tests $v_{a\restriction n}$ converge to δ_a; along an antichain they converge to $\mathbf{0}$. Hence the extended Alexandroff compactification is separable but not first countable.

(3) The *split Cantor set* $S(2^{\mathbb{N}}) = \{\chi_{[a}, \chi_{(a} : a \in 2^{\mathbb{N}}\}$. The continuous thresholds $\chi_{[t^\frown 0^\infty}$, together with the isolated point $\mathbf{0} = \chi_{(1^\infty}$, form a countable dense subset: thresholds approaching a from below converge to $\chi_{[a}$, and from above to $\chi_{(a}$. It follows that the split Cantor set is separable, first countable and hereditarily separable, but not metrizable.

(4) The *Alexandroff duplicate* $D(2^{\mathbb{N}}) = \{(\mathbf{0}, a), (\delta_a, a) : a \in 2^{\mathbb{N}}\} \subseteq A(2^{\mathbb{N}}) \times 2^{\mathbb{N}}$. The points (δ_a, a) are isolated, and every neighborhood of $(\mathbf{0}, a)$ contains, for all b in some neighborhood of a, the point $(\mathbf{0}, b)$ and, with finitely many exceptions, the point (δ_b, b). Consequently, the Alexandroff duplicate is first countable but not separable; it is contained in the separable compactum $\hat{D}(2^{\mathbb{N}}) = \overline{\{(v_t, t^\frown 0^\infty) : t \in 2^{<\mathbb{N}}\}} = D(2^{\mathbb{N}}) \cup \{(v_t, t^\frown 0^\infty) : t \in 2^{<\mathbb{N}}\}$.

(5) The *extended duplicate of the split Cantor set*

$$\hat{D}(S(2^{\mathbb{N}})) = \overline{\{(v_t, \chi_{[t^\frown 0^\infty})\}_t} \subseteq \hat{A}(2^{\mathbb{N}}) \times S(2^{\mathbb{N}}),$$

which consists of the points $(v_t, \chi_{[t^\frown 0^\infty})$ and, for $a \in 2^{\mathbb{N}}$, the points $(\delta_a, \chi_{[a})$, $(\mathbf{0}, \chi_{[a})$ and $(\mathbf{0}, \chi_{(a})$ [ADK08, 4.3.7]. Since the sequences $(a\restriction n)^\frown 0^\infty$ increase to a, the points $(v_{a\restriction n}, \chi_{[(a\restriction n)^\frown 0^\infty})$ converge to $(\delta_a, \chi_{[a})$ as $n \to \infty$. The extended duplicate of the split Cantor set is therefore separable and first countable; since the points $(\delta_a, \chi_{[a})$ form an uncountable discrete subspace, it is not hereditarily separable.

The constructions of the Alexandroff compactification and of the Alexandroff duplicate make sense with $2^{\mathbb{N}}$ replaced by any compact metric space K; the resulting spaces are denoted $A(K)$ and $D(K)$, respectively. Since $A(K)$ is the one-point compactification of a discrete space of the cardinality of K, it depends only on that cardinality; for instance, $A([0,1])$ is homeomorphic to $A(2^{\mathbb{N}})$.

1.5. **NIP in model theory, combinatorics and learning: polynomial versus exponential growth.** We now trace the NIP of Definition 1.5 to its counterparts in model theory, combinatorics and statistical learning, and make precise how it is related to them; the relation is used in Section 5, where a bound on a Vapnik–Chervonenkis dimension yields the NIP of Bresenham's algorithm. For families of $\{0,1\}$-valued functions, that is, of sets, a combinatorial dividing line closely related to the NIP appeared in 1971–1972, some years before Theorem 1.4, independently and almost simultaneously in three fields: in model theory, in the work of Saharon Shelah [She71, She90]; in combinatorics, in the work of Norbert Sauer [Sau72] and, again, of Shelah [She72, She90]; and in statistical learning, in the work of Vladimir Vapnik and Alexey Chervonenkis [VC71, VC74]. In each of them, the line separates the "tame" families of sets, whose number of traces on an n-element set (the intersections of the members of the family with the set) is bounded by a polynomial in n, from the "wild" ones, for which, for every $n \in \mathbb{N}$, some n-element set has all 2^n possible traces.

In model theory: Among the dividing lines of Shelah's classification theory [She90], the independence property is detected by counting types over finite sets of parameters. Let T be a complete first-order theory, let $\varphi(x;y)$ be a formula of its language, in which x and y are finite tuples of variables of lengths $|x|$ and $|y|$, and let B be a finite set of $|y|$-tuples of elements of a model M of T. A *φ-type* over B is a maximal set of formulas $\varphi(x;b)$ and $\neg\varphi(x;b)$, with $b \in B$, that is consistent with the theory of M with constants for the elements of M; since B is finite, every φ-type over B is realized by an $|x|$-tuple a of elements of M, and it is determined by the set $\{b \in B : M \models \varphi(a;b)\}$. Hence the set $S_\varphi(B)$ of φ-types over B is in bijection with the set of traces on B of the family of sets $\{b : M \models \varphi(a;b)\}$, with a ranging over $|x|$-tuples of elements of M, and $|S_\varphi(B)| \leq 2^{|B|}$. The formula φ has the *independence property* (IP) if, for every $n \in \mathbb{N}$, some set B of n tuples satisfies $|S_\varphi(B)| = 2^n$; otherwise the formula is *NIP*, or *dependent*, and the theory T is NIP if all its formulas are NIP. Since T is complete, these notions do not depend on the model M.

If φ is NIP, the Sauer–Shelah lemma below bounds $|S_\varphi(B)|$ by a polynomial in $|B|$. If φ has IP, then for every n some set A of at most $n|y|$ parameters satisfies $|S_\varphi(A)| \geq 2^n$, where $S_\varphi(A)$ denotes the set of φ-types over the set of $|y|$-tuples of elements of A [Sim15, Section 2.3.4]. Consequently, φ is NIP if and only if $|S_\varphi(A)|$ is bounded by a polynomial in $|A|$ as A ranges over the finite sets of parameters. Over infinite sets of parameters, counting φ-types does not separate NIP from IP without further set-theoretic hypotheses: if κ is infinite and $2^\kappa = \kappa^+$, every unstable formula, NIP or not, already has 2^κ φ-types over some set of κ parameters [Sim15, Section 2.3.4].

In combinatorics: Sauer [Sau72] and Shelah [She72] proved independently the following dichotomy, now known as the *Sauer–Shelah lemma*. Let $\mathscr{F} = \{S_i : i \in I\}$ be a family of subsets of a set S, indexed by a set I. Then either for every $n \in \mathbb{N}$ there is a set $A \subseteq S$ with $|A| = n$ such that $|\{S_i \cap A : i \in I\}| = 2^n$, in which case $\mathscr{F}$ is said to have "exponential complexity", or there exists $N \in \mathbb{N}$ such that, for every $A \subseteq S$ with $|A| \geq N$,

$$|\{S_i \cap A : i \in I\}| \leq \sum_{j=0}^{N-1} \binom{|A|}{j},$$

in which case $\mathscr{F}$ is said to have "polynomial complexity" (every family of subsets of a *finite* set S has polynomial complexity). The name is justified by the following estimate, valid for fixed $N > 0$ and for $|A| = m > 0$, which bounds the number of traces by a polynomial in m:

$$\sum_{j=0}^{N-1} \binom{m}{j} \leq \sum_{j=0}^{N-1} \frac{m^j}{j!} \leq \left(\sum_{j=0}^{N-1} \frac{1}{j!}\right) \cdot m^{N-1} < e \cdot m^{N-1}.$$

In machine learning: The Sauer–Shelah lemma is also the dichotomy that Vapnik and Chervonenkis [VC71, VC74] had discovered and proved slightly earlier, in 1971, in their study of uniform convergence in statistics. The largest size of a set $A \subseteq S$ with $2^{|A|}$ traces $S_i \cap A$, that is, the integer $N-1$ for the least integer N as in the Sauer–Shelah lemma, is called the

VC dimension of $\mathscr{F}$; if no such integer N exists, the VC dimension of $\mathscr{F}$ is infinite. Through the uniform convergence theorem of Vapnik and Chervonenkis, the lemma bounds the number of data points (the sample size) needed to "learn" a concept class $\mathscr{F}$ of finite VC dimension in a statistical sense, that is, up to a preset error and with a preset confidence. The Fundamental Theorem of Statistical Learning states that a class of $\{0,1\}$-valued functions is PAC-learnable (PAC stands for "Probably Approximately Correct" [Val84]) if and only if its VC dimension is finite, provided that the class satisfies a mild measurability condition [BEHW89]; see also [SSBD14, Theorem 6.7].

For first-order formulas the three dichotomies coincide: by the description of φ-types given above, a formula $\varphi(x;y)$ has IP exactly when the family of sets $\{b : M \models \varphi(a;b)\}$, with a ranging over the $|x|$-tuples of elements of a model M of T, has infinite VC dimension, and for an NIP formula the Sauer–Shelah lemma gives the polynomial bound on the number of φ-types. For an arbitrary family of sets, finiteness of the VC dimension implies that the characteristic functions of its members have the NIP of Definition 1.5. The NIP does not, however, imply finite VC dimension: the finite subsets of $\mathbb{N}$ form a family of infinite VC dimension whose characteristic functions have the NIP.

Historical remarks. The Baire hierarchy of functions was introduced by René-Louis Baire in his 1899 doctoral thesis, *Sur les fonctions de variables réelles*. The theory of spaces of continuous functions with the topology of pointwise convergence, C_p-theory, was pioneered by A. V. Arhangel'skiĭ and his students in the 1970s and 1980s [Ark92], and it has since found numerous applications in model theory and in functional analysis; Tkachuk's problem book [Tka11] develops the subject systematically, through problems with complete solutions.

2. Compositional Computation Structures: A structural approach to arbitrary-precision computations

The theory of function spaces assembled in Section 1 is brought to bear on computation through the framework of compositional computation structures, in which deep computations become points of a function space; the present section is devoted to that framework. A computation on real-valued data is observed through its features, each known to finite precision, and a deep computation is a limit of computations in the topology of this observation, the topology of pointwise convergence. We first recall the framework of the first paper of this series [ADIW26]: states are described by their types, the tuples of values of all their features; computations act on states and are composed; and deep computations are the pointwise limits of computations, taken in the space of types. We then recall the stable sets of computations, studied in that paper, and close the section with two examples that motivate the classification of Section 3.

2.1. Computation States Structures.

The first ingredient of the framework, which we introduce here, is the set of states on which computations act, together with the real-valued features through which the states are observed. A *computation states structure* is a pair $(L, \mathcal{P})$, where L is a set whose elements we call *states* and $\mathcal{P}$ is a collection of real-valued functions on L that we call *predicates* or *features*. Intuitively, L is the set of states of a computation, and the predicates are the

primitives that are given and accepted as computable; for $P \in \mathcal{P}$ and $v \in L$, the number $P(v)$ is the value of the (observable) feature P at the state v. Maps $f\colon L \to L$ are called *transitions* of $(L, \mathcal{P})$; the computations of Definition 2.4 below are transitions.

The *type* of a state $v \in L$ is the indexed family[4]

$$\mathrm{tp}(v) = (P(v))_{P\in\mathcal{P}} \in \mathbb{R}^{\mathcal{P}}$$

of its features.

We assume that each state $v \in L$ is determined by its type $\mathrm{tp}(v)$; we may then identify L with its image $\mathrm{tp}[L] \subseteq \mathbb{R}^{\mathcal{P}}$ under the type map and give it the subspace topology, that is, the topology of pointwise convergence induced by the product topology of $\mathbb{R}^{\mathcal{P}}$. In particular, for each $P \in \mathcal{P}$, the map $v \mapsto P(v)$ is continuous on L.

The principal examples are the state space $L = \mathbb{R}^{\mathbb{N}}$, with the predicates $P_i(v) = v_i$ for $i \in \mathbb{N}$, and, for a positive integer n, the state space $L = \mathbb{R}^n$, with the predicates $P_i(v) = v_i$ for $0 \le i < n$. Their subspaces are the state spaces of primary interest in this paper; through the identification of each state with its type, every computation states structure with countably many predicates is, once its predicates are enumerated, such a subspace.

Definition 2.1. Let $(L, \mathcal{P})$ be a computation states structure. The type $\mathrm{tp}(v) \in \mathbb{R}^{\mathcal{P}}$ of a state $v \in L$ is called a *realized* (state) type. The closure of the set of realized types in $\mathbb{R}^{\mathcal{P}}$, with the topology of pointwise convergence, is called the *space of (state) types* of $(L, \mathcal{P})$ and is denoted $\mathcal{L}$; its elements $\mathbf{v} \in \mathcal{L}$ are called *state types*, and those in $\mathcal{L} \setminus L$ are called *unrealized* state types.

State types formalize the intuitive notion of a limit state, and the predicates extend to them continuously. Indeed, for each $P \in \mathcal{P}$, the P-th coordinate map $\mathbb{R}^{\mathcal{P}} \to \mathbb{R}$, $\mathbf{v} = (\mathbf{v}_Q)_{Q\in\mathcal{P}} \mapsto \mathbf{v}_P$, restricts to a continuous extension to $\mathcal{L}$ of the predicate $P(\cdot)$ on L; abusing notation, we still denote this extension by $P(\cdot)$, or just P.

Remark 2.2. One may regard unrealized state types as "hyper-states"; when such a type is obtained as the limit of a sequence (or as an ultralimit) of realized state types, it may be called a "deep state". For readers versed in model theory we note that, at least for the purpose of realizing state types with parameters in the state space L, the type space $(\mathcal{L}, \mathcal{P})$ serves as a "poor man's" ultraproduct of $(L, \mathcal{P})$ in which, by the very definition of $\mathcal{L}$, every type $\mathbf{v}$ is realized.

2.1.1. *Bounds on precision and magnitude. Shards.* The analysis of this paper takes place on shards, the sets of states on which each feature obeys a prescribed bound, and before defining them we explain why actual computation imposes such bounds. Computation states structures are designed to model actual computation, in which operations on real-valued quantities can be carried out only approximately, to a

[4]As in Proposition 1.8, where $\mathcal{P}$ is merely a set of symbols, we regard each predicate as named by a formal label P, a *predicate symbol*, whose meaning in the computation states structure is the real function $P(\cdot)\colon v \mapsto P(v)$ on L. The symbol P thus names an observable feature, whose value at a state $v \in L$ is $P(v)$. Whenever $\mathcal{P}$ is used as an index set, as in the product $\mathbb{R}^{\mathcal{P}}$ below, it is regarded as a set of symbols. The distinction between a symbol and its meaning is usually suppressed in mathematics, but the model-theoretic framework requires awareness of it. When necessary, we write $P(\cdot)$ for the function on L and P for the symbol; when no confusion can arise, we also write P for the function.

preset degree of precision. A bound on the magnitude of the quantities involved is no less important in practice, for no machine can store, let alone compute with, arbitrarily large numbers. An actual computation therefore fixes at the outset bounds for its errors and for the absolute magnitudes of the quantities it handles, even when the collection $\mathcal{P}$ of predicates is infinite, possibly uncountable; different parts of a computation may require different error bounds and may deal with quantities of very different sizes. Focusing on magnitudes, we consider sets of computations for which a bound for each predicate is fixed in advance, and we are thereby led to the notion of *shard*, introduced in the first paper of this series [ADIW26].

Definition 2.3. Let $(L, \mathcal{P})$ be a computation states structure, with L identified with the set $\mathrm{tp}[L] \subseteq \mathbb{R}^{\mathcal{P}}$ of its realized types. A *sizer* is a family $r_\bullet = (r_P)_{P\in\mathcal{P}}$ of positive real numbers indexed by $\mathcal{P}$. For a sizer $r_\bullet$, let $\mathbb{R}[r_\bullet] \coloneqq \prod_{P\in\mathcal{P}}[-r_P, r_P]$, a compact space, and define the $r_\bullet$-*shard* of the state space L by

$$L[r_\bullet] = L \cap \mathbb{R}[r_\bullet].$$

The $r_\bullet$-*type shard* is the closure $\mathcal{L}[r_\bullet] = \overline{L[r_\bullet]}$, a closed and hence compact subset of $\mathbb{R}[r_\bullet]$.

The union of all type shards, as the sizer $r_\bullet$ varies, is denoted $\mathcal{L}_{\mathrm{sh}}$ and called the space of *shard-supported types*.

We have $\mathcal{L}_{\mathrm{sh}} \subseteq \mathcal{L}$, and the inclusion may be proper; equality holds in the important special case when $\mathcal{P}$ is countable (see [ADIW26]).

2.2. **Compositional Computation Structures.** The compositional computation structures of the first paper of this series [ADIW26], within which deep computations are defined, are recalled in this subsection, beginning with the ingredients of their definition. Computations transform states, and two computations performed one after the other form a computation; the computations of a system therefore form a semigroup under composition, which we take to contain the identity, the computation that does nothing. Deep computations are the limits of computations in the topology of pointwise convergence, taken in the space of types. Since an actual computation keeps the magnitudes that it handles within bounds, the shards of Section 2.1.1 enter the definition through the notion of confinement. The following definition records these ingredients.

Definition 2.4. A *compositional computation structure* (CCS) is a triple $(L, \mathcal{P}, \Gamma)$, where

- $(L, \mathcal{P})$ is a computation states structure, and
- $\Gamma \subseteq L^L$ is a semigroup under composition that contains the identity map id of L, that is, a monoid of transformations of L.

The elements of Γ are called the *computations* of $(L, \mathcal{P}, \Gamma)$. The state space L is identified with a subset of the type space $\mathcal{L} \subseteq \mathbb{R}^{\mathcal{P}}$ of $(L, \mathcal{P})$.

We topologize Γ as a subset of the product L^L, in which, as usual, the exponent L serves merely as an index set, while the base $L \subseteq \mathbb{R}^{\mathcal{P}}$ carries the topology of pointwise convergence. The semigroup Γ is thus a subset of the product $\mathcal{L}^L \subseteq (\mathbb{R}^{\mathcal{P}})^L$, and its closure $\overline{\Gamma}$ is contained in $\mathcal{L}^L$. The elements $\xi \in \overline{\Gamma}$ are called (real-valued) *deep computations* or *ultracomputations*.

A collection R of sizers is *exhaustive* if $L = \bigcup_{r_\bullet\in R} L[r_\bullet]$. A computation $\gamma \in \Gamma$ is *R-confined* if it maps $L[r_\bullet]$ into itself for every $r_\bullet \in R$, and a subset $\Delta \subseteq \Gamma$ is *R-confined* if each $\gamma \in \Delta$ is.

Unlike its subspace L^L, which is a semigroup under composition, the space $\mathcal{L}^L$ has no natural semigroup structure; deep computations are therefore not composable in general.

Proposition 2.5. *If $\Delta \subseteq \Gamma$ is confined by an exhaustive collection of sizers, then its closure $\overline{\Delta}$ is a compact subset of $\mathcal{L}^L_{\mathrm{sh}}$.*

Proof. Let R be an exhaustive collection of sizers that confines Δ, and choose, for each $v \in L$, a sizer $r_\bullet^{(v)} \in R$ with $v \in L[r_\bullet^{(v)}]$. Since every $\gamma \in \Delta$ maps v into $L[r_\bullet^{(v)}]$, the set Δ lies in the compact space $\prod_{v\in L} \mathcal{L}[r_\bullet^{(v)}] \subseteq \mathcal{L}^L_{\mathrm{sh}}$, which therefore contains $\overline{\Delta}$. □

For a CCS $(L, \mathcal{P}, \Gamma)$, we regard the elements of Γ as "standard" finitary computations, and the elements of $\overline{\Gamma}$, i.e., deep computations, as possibly infinitary limits of standard computations.

Remarks 2.6. (For readers versed in model theory.)[5]

(1) Compositional computation structures, as defined above, are not literally real-valued structures, because nothing in the structure itself forces Γ to be realized as a subset of the product space L^L. However, as in their original definition [ADIW26], they may be regarded as real-valued structures $(L, \mathcal{P}, \Gamma, \circ, \mathrm{ev})$, where
   - $(L, \mathcal{P})$ is a computation states structure,
   - $(\Gamma, \circ)$ is a semigroup,
   - $\mathrm{ev} : \Gamma \times L \to L : (\gamma, v) \mapsto \mathrm{ev}(\gamma, v)$ is a semigroup action of Γ on L (i.e., $\mathrm{ev}(\gamma \circ \delta, v) = \mathrm{ev}(\gamma, \mathrm{ev}(\delta, v))$ for $\gamma, \delta \in \Gamma$ and $v \in L$).

   The two definitions are reconciled by identifying each $\gamma \in \Gamma$ with the map $v \mapsto \mathrm{ev}(\gamma, v)$.

(2) Regarded as real-valued structures in this sense, the compositional computation structures with a fixed collection $\mathcal{P}$ of predicate symbols form an elementary class. The notion of state type introduced above (essentially, that of a quantifier-free type with parameters of sort L only) may be enriched by further formulas, for instance with parameters in Γ or with quantifiers, and extended to types of sort Γ ("transform-types"). As usual, all types over a structure $\underline{C} = (L, \mathcal{P}, \Gamma, \circ, \mathrm{ev})$ are realized in some ultrapower $\underline{C}^{(\mathcal{U})}$ of $\underline{C}$, constructed as in the references above.[6] In Banach space theory, this approach to spaces of types was introduced by Krivine [Kri74, KM81, Kri72].

2.3. **Computability of deep computations and the Extendibility Axiom.** The computability of transforms and the Extendibility Axiom, which requires every computation to have an extension that is continuous on each type shard, are the two notions on which the rest of the paper depends, and we recall them in this subsection. A machine evaluates only sums and products of finitely many observed features, and only to within a preset error on a preset range of magnitudes; for this reason, computability is defined by uniform approximation on shards by polynomials in the features. Let $(L, \mathcal{P})$ be a computation states structure. A map

[5]The real-valued structures used here are those of Keisler's model theory for real-valued structures [Kei23]; see also [DnI17, Section 6] for an introduction.

[6]By contrast, no ultrapower is needed to realize state types in a computation states structure (see Remark 2.2).

$f\colon L \to \mathcal{L}$ from its state space to its space of types is called a *transform*. In the two definitions that follow, and in the characterization of computability after them, the collection $\mathcal{P}$ of predicates is assumed to be *countable*.

(1) A *computable predicate* is a function $\varphi\colon L \to \mathbb{R}$ whose restriction to each shard is a uniform limit of polynomials, with real coefficients, in the features $P(\cdot)$.
(2) A transform $f\colon L \to \mathcal{L}$ is *computable* if, for each $Q \in \mathcal{P}$, the output feature $Q \circ f\colon L \to \mathbb{R}$ is a computable predicate.

Alva, Dueñez, Iovino and Walton [ADIW26] showed that a transform $f\colon L \to \mathcal{L}$ is computable if and only if it extends to a continuous map $\tilde{f}\colon \mathcal{L} \to \mathcal{L}$. This characterization motivates the following definition, in which continuity is required on each type shard.

A computation $\gamma \in \Gamma$ is *extendable* if it has an extension $\tilde{\gamma}\colon \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ such that, for every sizer $r_\bullet$, there is a sizer $s_\bullet$ for which the restriction $\tilde{\gamma}|_{\mathcal{L}[r_\bullet]}\colon \mathcal{L}[r_\bullet] \to \mathcal{L}[s_\bullet]$ is continuous. Such an extension is unique, since it is continuous on each type shard $\mathcal{L}[r_\bullet]$, in which $L[r_\bullet]$ is dense. For a set $\Delta \subseteq \Gamma$ of extendable computations, we write $\tilde{\Delta}$ for the set $\{\tilde{\gamma}\colon \gamma \in \Delta\}$.

Extendibility Axiom. *Every computation $\gamma \in \Gamma$ is extendable.*

The Extendibility Axiom is discussed at length in the first paper of this series [ADIW26]. *From now on, all compositional computation structures are assumed to satisfy the Extendibility Axiom.*

The following lemma extends deep computations to the type shards; it is used in Theorems 3.1 and 6.7.

Lemma 2.7. *Let R be an exhaustive collection of sizers and let $\Delta \subseteq \Gamma$ be R-confined. Every deep computation $f \in \overline{\Delta}$ has an extension $\tilde{f}\colon \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ that, for every $r_\bullet \in R$, maps $\mathcal{L}[r_\bullet]$ into itself and satisfies $\tilde{f}|_{\mathcal{L}[r_\bullet]} \in \overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$.*

Proof. Write $f = \lim_{i,\mathcal{U}} \gamma_i$, where $(\gamma_i)_{i \in I}$ is a family in Δ and $\mathcal{U}$ is an ultrafilter on the index set I, and let $r_\bullet \in R$ be given. Each $\tilde{\gamma}_i$ maps $L[r_\bullet]$ into itself and is continuous on $\mathcal{L}[r_\bullet]$, so it maps the closure $\mathcal{L}[r_\bullet]$ of $L[r_\bullet]$ into itself; since $\mathcal{L}[r_\bullet]$ is compact, the limit $\tilde{f}(v) \coloneqq \lim_{i,\mathcal{U}} \tilde{\gamma}_i(v)$ exists for every $v \in \mathcal{L}[r_\bullet]$ and lies in $\mathcal{L}[r_\bullet]$, and $\tilde{f}|_{\mathcal{L}[r_\bullet]} = \lim_{i,\mathcal{U}} \tilde{\gamma}_i|_{\mathcal{L}[r_\bullet]}$. Since R is exhaustive, this defines $\tilde{f}$ on a set that contains L, and there $\tilde{f}$ agrees with f; at the remaining points of $\mathcal{L}_{\mathrm{sh}}$ we let $\tilde{f}$ be the identity. □

2.4. **Stability.** The highest level of the hierarchy studied in this paper is occupied by the stable sets of computations, all of whose deep computations are computable, and we recall them before descending, in Section 3, to the levels below. By the theorem of Alva, Dueñez, Iovino and Walton recalled above, the computable transforms are exactly those that extend to continuous maps on the space of types, and accordingly the stable sets are defined by the continuity of their deep computations on every type shard.

Definition 2.8. A set $\Delta \subseteq \Gamma$ of computations is *stable* if, for every sizer $r_\bullet$, every deep computation in $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is continuous, that is, computable.

Stable sets of computations are the subject of the first paper of this series [ADIW26], where they are characterized by Grothendieck's double limit criterion and identified with stability in the sense of model theory.

2.5. **Two motivating examples.** Before developing the classification, we describe two computations whose deep computations differ sharply in character, for the contrast between them indicates what the classification must distinguish; both are treated in detail in Section 4.

Square roots. Newton's method for $p(x) = x^2 - a$, with $a > 0$, acts on the real projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$ by the map $N_p(x) = (x^2 + a)/(2x)$, with $N_p(0) = N_p(\infty) = \infty$. Its iterates are continuous, and they converge pointwise to the map $\mathfrak{f}$ that sends every positive number to $\sqrt{a}$, every negative number to $-\sqrt{a}$, and both 0 and ∞ to ∞ (see Figure 1). The deep computations are the iterates together with $\mathfrak{f}$. Since $\mathfrak{f}$ is discontinuous, it is not computable, and the set of iterates is not stable; nevertheless, the map $\mathfrak{f}$ is the limit of a *sequence* of computations, and the space of deep computations is countable.

Newton's method on the Riemann sphere. For $\rho(z) = z^3 - 2z + 2$, Newton's method on the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ displays three distinct modes of behavior, each confined to its own part of the sphere (see Figure 2 in Section 4.1). On the basins of the three roots the iterates converge; on the basin of the cycle $0 \mapsto 1 \mapsto 0$ they do not, although the even iterates and the odd iterates converge there separately, to different limits; and on the common boundary of these basins, a fractal set, the iterates are chaotic. For the simpler polynomial $z^2 - 1$, chaos of this kind produces deep computations that are not even limits of sequences of computations (see Proposition 5.10 and Theorem 3.1).

The classification of Section 3 is designed to separate such behaviors. In Theorem 3.1 below we characterize the sets of computations whose deep computations are of the first Baire class on every type shard; for them, as in the case of square roots, every deep computation is, on each shard, the limit of a sequence of computations. Todorčević's trichotomy, which we recall in Section 3.2, then measures the complexity of the space of such deep computations.

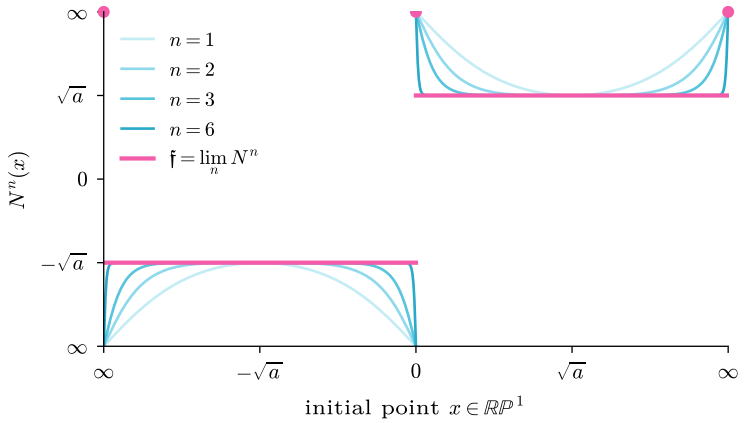


FIGURE 1. Newton iterates N_p^n for $p(x) = x^2 - a$, with $a > 0$, on the real projective line, drawn as graphs whose axes run from ∞ through 0 back to ∞. The continuous maps N_p^n converge pointwise to the deep computation $\mathfrak{f}$, which sends the positive half line to $\sqrt{a}$, the negative half line to $-\sqrt{a}$, and 0 and ∞ to ∞.

3. Classifying deep computations

The main theorems of the paper, which classify sets of computations by the complexity of their deep computations and thereby yield a new hierarchy of complexity classes, are proved in this section by means of the topology of function spaces. The examples of Section 2.5 call for such a classification, since some natural algorithms have discontinuous deep computations that are nevertheless limits of sequences of computations, while others have deep computations that are not even limits of sequences. The main topological tool is Todorčević's trichotomy for Rosenthal compacta [Tod99], and our model is the work of Glasner and Megrelishvili on tame semigroups in topological dynamics [GM22].

3.1. NIP, Rosenthal compacta, and deep computations.

The classification rests on Theorem 3.1, which we prove in this subsection and which characterizes, by a finitary condition, the sets of computations whose deep computations are of the first Baire class on every type shard, just as the first paper of this series [ADIW26] characterized, by Grothendieck's double limit criterion, those whose deep computations are continuous.

Theorem 3.1. *Let $(L, \mathcal{P}, \Gamma)$ be a compositional computation structure (see Definition 2.4) satisfying the Extendibility Axiom, with $\mathcal{P}$ countable. Let R be an exhaustive collection of sizers, and suppose that $\Delta \subseteq \Gamma$ is R-confined. Then the following conditions are equivalent:*

(i) $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ *for all $r_\bullet \in R$;*

(ii) $\pi_P \circ \Delta|_{L[r_\bullet]}$ *satisfies the NIP for all $P \in \mathcal{P}$ and $r_\bullet \in R$, where $\pi_P \colon \mathbb{R}^{\mathcal{P}} \to \mathbb{R}$ is the projection onto the coordinate P; that is, for all $P \in \mathcal{P}$, $r_\bullet \in R$, reals $a < b$ and sequences $(\gamma_n)_{n\in\mathbb{N}}$ in Δ, there are finite disjoint sets $E, F \subseteq \mathbb{N}$ such that*

$$L[r_\bullet] \cap \bigcap_{n\in E} (\pi_P \circ \gamma_n)^{-1}(-\infty, a] \cap \bigcap_{n\in F} (\pi_P \circ \gamma_n)^{-1}[b, \infty) = \emptyset.$$

Furthermore, if these conditions hold, then every deep computation $f \in \overline{\Delta}$ has an extension $\tilde{f} \colon \mathcal{L}_{sh} \to \mathcal{L}_{sh}$ such that $\tilde{f}|_{\mathcal{L}[r_\bullet]} \in \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ for all $r_\bullet \in R$. In particular, on each shard, every deep computation is the pointwise limit of a sequence of computations.

Proof. Since $\mathcal{P}$ is countable, each type shard $\mathcal{L}[r_\bullet]$ is a compact metrizable, hence Polish, subspace of $\mathbb{R}^{\mathcal{P}}$. Let $r_\bullet \in R$ be given. By the Extendibility Axiom and confinement, every $\tilde{\gamma}$ with $\gamma \in \Delta$ maps $\mathcal{L}[r_\bullet]$ continuously into itself (see the proof of Lemma 2.7), and hence $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ is a pointwise bounded set of continuous functions for every $P \in \mathcal{P}$. By Theorem 1.9 and Lemma 1.10, condition (ii) holds at $r_\bullet$ if and only if $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathbb{R}^{\mathcal{P}})$, which (i) implies. Conversely, suppose that this inclusion holds. For a Polish space X, every compact subset of $\mathrm{B}_1(X)$ is Fréchet–Urysohn, that is, every point in the closure of a subset is the limit of a sequence of points of the subset (see [BFT78, Theorem 3F] or [Deb13, Theorem 4.1]). Applied to $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$, regarded by Proposition 1.8 as a compact subset of $\mathrm{B}_1(\mathcal{L}[r_\bullet] \times \mathcal{P})$, where $\mathcal{L}[r_\bullet] \times \mathcal{P}$ is Polish because $\mathcal{P}$ is countable and discrete, this shows that every member of $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is the pointwise limit of a sequence of maps $\tilde{\gamma}_n|_{\mathcal{L}[r_\bullet]}$ with $\gamma_n \in \Delta$, which are continuous maps from $\mathcal{L}[r_\bullet]$ to itself; hence (i) holds at $r_\bullet$. Finally, if (i)

holds, the extension $\tilde{f}$ of Lemma 2.7 satisfies $\tilde{f}|_{\mathcal{L}[r_\bullet]} \in \overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ for all $r_\bullet \in R$, and, by the argument just given, every deep computation is, on each shard, the pointwise limit of a sequence of computations. □

3.2. **Todorčević's trichotomy and levels of NIP.** Todorčević's trichotomy now enables us to stratify the NIP sets of computations into three levels. The trichotomy concerns separable Rosenthal compacta, and, by Theorem 3.1 and under its hypotheses, the deep computations of a countable NIP set of computations form such a compactum on each type shard. We measure the complexity of such a compactum by three topological properties, metrizability, hereditary separability and first countability, which form a chain: every compact metrizable space is hereditarily separable, and R. Pol proved that every hereditarily separable Rosenthal compactum is first countable (see [Deb13, Section 10]). These properties define three levels of NIP.

Definition 3.2. Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom, with $\mathcal{P}$ countable, and let R be an exhaustive collection of sizers. Let $\Delta \subseteq \Gamma$ be an R-confined countable set of computations satisfying the NIP on shards and features (condition (ii) in Theorem 3.1). We say that Δ is:

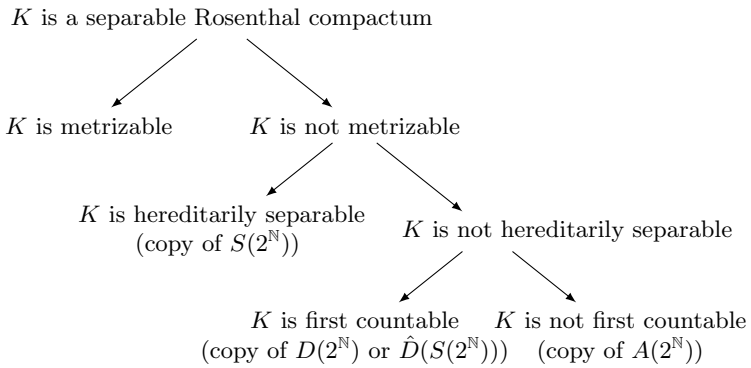
(i) NIP_1 if $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is first countable for every $r_\bullet \in R$;
(ii) NIP_2 if $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is hereditarily separable for every $r_\bullet \in R$;
(iii) NIP_3 if $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is metrizable for every $r_\bullet \in R$.

By the chain of properties above, the levels satisfy $\mathrm{NIP}_3 \Rightarrow \mathrm{NIP}_2 \Rightarrow \mathrm{NIP}_1 \Rightarrow$ NIP. Moreover, a countable stable set of computations (see Definition 2.8), which has the NIP by Theorem 3.1, is NIP_3: on each shard its deep computations form a separable compact subset of a countable product of spaces $\mathrm{C_p}(\mathcal{L}[r_\bullet])$, that is, a separable Eberlein compactum, and separable Eberlein compacta are metrizable [Ark92]. The five compacta of Section 1.4 show that none of the implications between the three topological properties can be reversed for separable Rosenthal compacta: the extended Alexandroff compactification $\hat{A}(2^{\mathbb{N}})$ is separable but not first countable; the extended duplicate $\hat{D}(S(2^{\mathbb{N}}))$ of the split Cantor set and the compactum $\hat{D}(2^{\mathbb{N}})$ that contains the Alexandroff duplicate are separable and first countable but not hereditarily separable; and the split Cantor set $S(2^{\mathbb{N}})$ is hereditarily separable but not metrizable. Todorčević proved that these examples are canonical [Tod99], in the following sense.

Theorem 3.3 (Todorčević's trichotomy [Tod99]; see also [ADK08, Theorem 3])**.** *Let K be a separable Rosenthal compactum.*

(i) If K is hereditarily separable but non-metrizable, then the split Cantor set $S(2^{\mathbb{N}})$ embeds into K.
(ii) If K is first countable but not hereditarily separable, then either the Alexandroff duplicate $D(2^{\mathbb{N}})$ or the extended duplicate $\hat{D}(S(2^{\mathbb{N}}))$ of the split Cantor set embeds into K.
(iii) If K is not first countable, then the Alexandroff compactification $A(2^{\mathbb{N}})$ embeds into K.

Together with the chain of properties above, the trichotomy yields the following classification of separable Rosenthal compacta:



The three levels admit a second description, in terms of the number of tests required to distinguish deep computations from one another. Let Δ and R be as in Definition 3.2, fix $r_\bullet \in R$, and write $K_{r_\bullet} = \overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$. We identify each $f \in K_{r_\bullet}$ with the real-valued function $(x, P) \mapsto P(f(x))$ on $\mathcal{L}[r_\bullet] \times \mathcal{P}$, whose points we call *tests*: the test (x, P) applies the feature P to the output on the input x. The topology of pointwise convergence on $K_{r_\bullet}$ is then the topology of pointwise convergence on tests. A set D of tests *separates* a set of deep computations if any two distinct members of the set differ on some test in D. A member f of $K_{r_\bullet}$ is *countably determined* if some countable set D_f of tests separates f from every other member of $K_{r_\bullet}$, and a set $E \subseteq K_{r_\bullet}$ is *finitely separated* if, for each $f \in E$, there are finitely many tests and a real number $\varepsilon_f > 0$ such that every $g \in E \setminus \{f\}$ differs from f by at least ε_f on one of these tests.

Proposition 3.4. *Let Δ and R be as in Definition 3.2, let $r_\bullet \in R$ be given, and let $K_{r_\bullet} = \overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$.*

(i) The compactum $K_{r_\bullet}$ is metrizable if and only if one countable set of tests separates $K_{r_\bullet}$.
(ii) It is first countable if and only if every member of $K_{r_\bullet}$ is countably determined.
(iii) It is hereditarily separable if and only if no finitely separated subset of $K_{r_\bullet}$ has cardinality $2^{\aleph_0}$.

Proof. (i) If a countable set D of tests separates $K_{r_\bullet}$, restriction to D is a continuous injection of the compact space $K_{r_\bullet}$ into the metrizable space $\mathbb{R}^D$, hence a homeomorphism onto its image. Conversely, if $K_{r_\bullet}$ is metrizable, then the square $K_{r_\bullet} \times K_{r_\bullet}$ is second countable, so the open set $\{(f, g) : f \neq g\}$ is Lindelöf; it is covered by the open sets $\{(f, g) : f(t) \neq g(t)\}$, for tests t, and a countable subcover gives the required set D. (ii) If f is countably determined, the singleton $\{f\}$ is a countable intersection of open sets, and in a compact Hausdorff space such a point has a countable base of neighborhoods. Conversely, let $(U_k)_{k\in\mathbb{N}}$ be a countable base of neighborhoods of f, and choose in each U_k a basic neighborhood of f defined by a finite set F_k of tests; then $D_f = \bigcup_k F_k$ separates f from every other member of $K_{r_\bullet}$. (iii) A set is finitely separated exactly when it is a discrete subspace, and, by Theorem 3.1, the compactum $K_{r_\bullet}$ is a separable Rosenthal compactum. A hereditarily separable space has no uncountable discrete subspace, since a separable discrete space is countable. Conversely, if $K_{r_\bullet}$ is not hereditarily separable, then, by Theorem 3.3, it contains a copy of $D(2^{\mathbb{N}})$, of $\hat{D}(S(2^{\mathbb{N}}))$ or of $A(2^{\mathbb{N}})$, and each of these contains a discrete subspace of cardinality $2^{\aleph_0}$ (see Section 1.4); for separable

Rosenthal compacta, the existence of such a subspace in the absence of hereditary separability is due to R. Pol (see [ADK08] and [GM22, Theorem 0.4]). □

Thus the set Δ is NIP_3 when, on each shard, a single countable set of tests separates all its deep computations; it is NIP_2 when no shard carries a finitely separated set of deep computations of the cardinality of the continuum; and it is NIP_1 when, on each shard, each deep computation is separated from all the others by countably many tests of its own.

In Section 5 we exhibit computations at each level of the hierarchy and show that every implication between the classes stable, NIP_3, NIP_2, NIP_1 and NIP is strict. The definitions of NIP_1, NIP_2 and NIP_3 are topological, and the descriptions of Proposition 3.4 still involve infinitely many tests; whether the three levels admit a finitary, combinatorial characterization, like that of NIP in Theorem 3.1, is Question 7.1.

3.3. **The Argyros–Dodos–Kanellopoulos heptachotomy and binary search.** For families of computations indexed by the binary tree, the heptachotomy of Argyros, Dodos and Kanellopoulos describes their closures more finely than Todorčević's trichotomy, which concerns arbitrary countable sets of computations, and we refine the classification accordingly, reading the minimal families of the heptachotomy as primitives of binary search. The countable dense subsets exhibited in the separable compacta of Section 1.4 are indexed by the binary tree $2^{<\mathbb{N}}$, and this indexing suits our computational emphasis: real numbers are represented by infinite binary sequences, which are the branches of the binary tree, and approximated by finite bitstrings, which are its nodes. A *dyadic subtree* of $2^{<\mathbb{N}}$ is a subset that is isomorphic to $2^{<\mathbb{N}}$ under the order of extension; we write it as $\{s_t : t \in 2^{<\mathbb{N}}\}$, indexed through the canonical isomorphism, which preserves extension and the lexicographic order.

Definition 3.5. Let X be a Polish space.

(1) If I is countable and $\{f_i : i \in I\} \subseteq \mathbb{R}^X$, $\{g_i : i \in I\} \subseteq \mathbb{R}^X$ are two families indexed by I, we say that they are *equivalent* if the map $f_i \mapsto g_i$ extends to a homeomorphism from $\overline{\{f_i : i \in I\}}$ to $\overline{\{g_i : i \in I\}}$.

(2) If $\{f_t : t \in 2^{<\mathbb{N}}\}$ is a pointwise bounded family, we say that it is *minimal* if, for every dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$, the family $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to $\{f_t : t \in 2^{<\mathbb{N}}\}$.

Argyros, Dodos and Kanellopoulos proved that, up to equivalence, there are exactly seven minimal families $\{f_t : t \in 2^{<\mathbb{N}}\}$ that are relatively compact in $\mathrm{B}_1(X)$, and that every relatively compact family $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq \mathrm{B}_1(X)$ contains, along a dyadic subtree, a family equivalent to one of them (see Theorem 3.6 below). We describe the seven minimal families next, in the notation of Argyros, Dodos and Kanellopoulos. A dyadic subtree is *regular* if each of its levels is contained in a level of $2^{<\mathbb{N}}$. Fix a regular dyadic subtree $T = \{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ such that $s^\frown 0^\infty \neq s'^\frown 0^\infty$ and $s^\frown 1^\infty \neq s'^\frown 1^\infty$ for all distinct $s, s' \in T$. The families are built from the prefix tests v_t and the threshold tests $\chi_{[a}$, $\chi_{(a}$ of Section 1.4. As in [ADK08], a point $\sigma \in 2^{\mathbb{N}}$, identified with a point of the Cantor set in $[0,1]$, is regarded as the constant function on $2^{\mathbb{N}}$ with value σ, and a pair (f, g) of functions on $2^{\mathbb{N}}$ is regarded as a single function on the disjoint union $2^{\mathbb{N}} \sqcup 2^{\mathbb{N}}$.

(1) $D_1 = \{\frac{1}{|t|+1} v_t : t \in 2^{<\mathbb{N}}\}$. Its closure is $D_1 \cup \{\mathbf{0}\}$, the one-point compactification of a countable discrete set.

(2) $D_2 = \{s_t^\frown 0^\infty : t \in 2^{<\mathbb{N}}\}$, a discrete subset of its closure $\overline{D_2}$, which Argyros, Dodos and Kanellopoulos denote by $2^{\leq\mathbb{N}}$; this closure is compact and metrizable, and its accumulation points form a copy of $2^{\mathbb{N}}$.
(3) $D_3 = \{\chi_{[s_t^\frown 0^\infty} : t \in 2^{<\mathbb{N}}\}$, a discrete subset of its closure $\overline{D_3}$, a closed subspace of $S(2^{\mathbb{N}})$.
(4) $D_4 = \{\chi_{(s_t^\frown 1^\infty} : t \in 2^{<\mathbb{N}}\}$, a discrete subset of its closure $\overline{D_4}$, a closed subspace of $S(2^{\mathbb{N}})$.
(5) $D_5 = \{v_t : t \in 2^{<\mathbb{N}}\}$, a discrete subset of its closure $\overline{D_5} = \hat{A}(2^{\mathbb{N}})$.
(6) $D_6 = \{(v_t, t^\frown 0^\infty) : t \in 2^{<\mathbb{N}}\}$, a discrete subset of its closure $\overline{D_6} = \hat{D}(2^{\mathbb{N}})$.
(7) $D_7 = \{(v_{s_t}, \chi_{[s_t^\frown 0^\infty}) : t \in 2^{<\mathbb{N}}\}$, a discrete subset of its closure $\overline{D_7}$, a closed subspace of $\hat{D}(S(2^{\mathbb{N}}))$.

Theorem 3.6 (Heptachotomy of minimal families [ADK08, Theorem 2])**.** *Let X be a Polish space. For every relatively compact family $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq \mathrm{B}_1(X)$, there exist $i \in \{1, 2, \ldots, 7\}$ and a regular dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ such that $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_i. Moreover, the families D_i are minimal and pairwise non-equivalent.*

For deep computations, the building blocks of the families $D_1, \ldots, D_7$ have a direct reading as primitives of binary search on $2^{\mathbb{N}}$, the space of binary expansions. A node $t \in 2^{<\mathbb{N}}$ determines the dyadic interval $[t^\frown 0^\infty, t^\frown 1^\infty]$, the set of inputs that round to t at precision $|t|$. The prefix test v_t asks whether the input rounds to t; the point $t^\frown 0^\infty$ is the rounded value itself; the comparisons $\chi_{[t^\frown 0^\infty}$ and $\chi_{(t^\frown 1^\infty}$ ask whether the input lies at or to the right of the left endpoint of the interval, or strictly to the right of its right endpoint. In Table 1 we read the seven families accordingly.

Family	Function at node t	Limits
D_1	prefix test weighted by $1/(\lvert t\rvert + 1)$	the test that never succeeds: confidence vanishes with depth
D_2	return the rounded value $t^\frown 0^\infty$	the exact input: successive approximation
D_3	compare with the left endpoint	comparison with an arbitrary real, ties broken upward
D_4	compare with the right endpoint	comparison with an arbitrary real, ties broken downward
D_5	prefix test	exact equality, or no match
D_6	prefix test and rounded value	a dictionary lookup that returns its key
D_7	prefix test and comparison	a three-way comparison: below, inside or above

TABLE 1. The minimal families of Argyros, Dodos and Kanellopoulos as primitives of binary search.

Let $\{\gamma_t : t \in 2^{<\mathbb{N}}\}$ be a family of computations of a CCS with countably many features, confined by an exhaustive collection R of sizers and satisfying the NIP on shards and features (condition (ii) of Theorem 3.1). By Theorem 3.1, for every feature P and every $r_\bullet \in R$, the functions $P \circ \tilde{\gamma}_t|_{\mathcal{L}[r_\bullet]}$ form a relatively compact

subset of $\mathrm{B}_1(\mathcal{L}[r_\bullet])$; by Theorem 3.6, this family contains, along a dyadic subtree, a family equivalent to one of the seven families $D_1, \dots, D_7$. In the reading of Table 1, every such family thus contains, on each shard and for each feature, one of the seven primitives of binary search, up to equivalence and along a dyadic subtree.

4. Examples of Compositional Computation Structures

From the general theory we turn to the examples that motivate it, among them the two announced in Section 2.5, and describe the deep computations of three compositional computation structures, namely Newton's method on the Riemann sphere for two cubic polynomials (Sections 4.1 and 4.2) and Newton's method for square roots on the real projective line (Section 4.3). In Section 5 we place the last of these, together with other classical algorithms, at the levels of the hierarchy of Section 3.

We begin by formalizing the Newton–Raphson root-finding method as a CCS [Gal00], [BFB16, Section 2.3]. Let $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ be the extended complex plane, that is, the Alexandroff (one-point) compactification of $\mathbb{C}$. We identify $\hat{\mathbb{C}}$ with the unit sphere $S^2 \subseteq \mathbb{R}^3$, the Riemann sphere, through the inverse stereographic projection, the homeomorphism $\hat{\mathbb{C}} \to S^2$ given by $z \mapsto (P_1(z), P_2(z), P_3(z))$, where $(P_1(\infty), P_2(\infty), P_3(\infty)) = (0, 0, 1)$ and, for $z \neq \infty$,

$$P_1(z) = \frac{2\,\Re z}{|z|^2 + 1}, \qquad P_2(z) = \frac{2\,\Im z}{|z|^2 + 1}, \qquad P_3(z) = \frac{|z|^2 - 1}{|z|^2 + 1}.$$

The extended complex plane $\hat{\mathbb{C}}$ thus carries the real predicates P_1, P_2, P_3, with values in $[-1, 1]$, and its topology is metrized by the pullback of the Euclidean metric of S^2.

Newton CCSs: steps and fractals. Here we describe the compositional computation structure generated by the Newton map, the single step that Newton's method iterates to approximate a root of a function, and recall the Fatou and Julia sets, on which the iterates of this map behave regularly and chaotically, respectively. The structure so obtained is the common setting of the three examples that follow. Let $p = p(z)$ be a nonconstant rational function with complex coefficients, that is, a nonconstant meromorphic map $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$. The *Newton step*, or *Newton map*, of p is the rational map

$$N_p : z \mapsto z - \frac{p(z)}{p'(z)}. \tag{4.1}$$

The iterates of the Newton map, that is, its compositional powers N_p^n, may be regarded as the computations of a CCS with state space $L = \hat{\mathbb{C}}$. The natural predicates are the coordinates P_1, P_2, P_3 of the inverse stereographic projection, since the real and imaginary parts $z \mapsto \Re z$ and $z \mapsto \Im z$ are not defined at $z = \infty$.

Let $\Gamma_p \coloneqq \{N_p^n : n \in \mathbb{N}\}$ be the semigroup of iterates of the Newton map (4.1). The CCS $(\hat{\mathbb{C}}, (P_1, P_2, P_3), \Gamma_p)$ is called *Newton's method for $p(z)$*, or the *Newton CCS for $p(z)$*. Since the state space $\hat{\mathbb{C}}$ is compact and the Newton map N_p is continuous on it, every Newton CCS satisfies the Extendibility Axiom.

Newton's method succeeds at an initial estimate $z_0 \in \hat{\mathbb{C}}$ when the sequence $(z_n)_{n \in \mathbb{N}}$ of Newton steps, where $z_n \coloneqq N_p^n(z_0)$, converges to a root of p.

Fatou and Julia sets. Recall that the *Fatou set* $\mathcal{F}_g$ of a rational map $g\colon \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ consists of the points that have a neighborhood on which every sequence of iterates of g has a subsequence that converges uniformly with respect to the spherical metric; its complement $\mathcal{J}_g := \hat{\mathbb{C}} \setminus \mathcal{F}_g$ is the *Julia set* of g, a closed set that is typically a fractal. The Julia set of the Newton map N_p is called the *Newton fractal* of p.

4.1. **Example: The Newton CCS for the cubic $\rho(z) = z^3 - 2z + 2$.** Our first example is the second motivating example of Section 2.5, a cubic polynomial whose Newton map has, besides the three roots, an attracting cycle. Newton's method therefore fails on an open set of initial estimates, its iterates do not converge, and among its deep computations are two distinct limits, one of the even iterates and one of the odd iterates. The polynomial $\rho(z) = z^3 - 2z + 2$ has Newton map

$$N_\rho : z \mapsto z - \frac{z^3 - 2z + 2}{3z^2 - 2} = \frac{2z^3 - 2}{3z^2 - 2}.$$

Denote the real root of ρ by r_0, and the roots in the upper and the lower half plane by r_+ and r_-. Starting from $z_0 = 0$, the Newton steps alternate $0, 1, 0, 1, \ldots$, and neither 0 nor 1 is a root of ρ.

Let B_0, B_+, B_- be the basins of the roots r_0, r_+, r_-, and let B_{01} be the basin of the superattracting cycle $\{0, 1\}$. The Fatou set of N_ρ is $B_{01} \cup B_0 \cup B_+ \cup B_-$, and each of these four sets is nonempty and open. Since B_{01} is nonempty, the iterates N_ρ^n do not converge pointwise. Let f_0 and f_1 be the deep computations obtained as limits of the iterates N_ρ^n along nonprincipal ultrafilters on $\mathbb{N}$ that contain the even and the odd numbers, respectively. On the Fatou set the even and the odd iterates converge pointwise, and there the two limits do not depend on the choice of the ultrafilters: both send each basin of a root to that root, while on B_{01} they take only the values 0 and 1 and exchange them, because $f_1 = N_\rho \circ f_0$ on the Fatou set by the continuity of N_ρ. In particular, the two deep computations are distinct.[7] Figure 2 depicts f_0 and f_1, together with the Newton fractal of ρ, on which the iterates behave chaotically.

4.2. **Example: The Newton CCS for the cubic $\kappa(z) = z^3 - 1$.** For contrast, we turn to $\kappa(z) = z^3 - 1$, whose roots are the cube roots of unity $r_0 = 1$ and $r_\pm = \exp(\pm 2\pi i/3)$ and whose Newton map is

$$N_\kappa : z \mapsto z - \frac{z^3 - 1}{3z^2} = \frac{2z^3 + 1}{3z^2}.$$

Unlike N_ρ, the map N_κ has no attracting cycle other than its three fixed points, the roots: the Fatou set is the union of the three basins, and on each basin the iterates converge to the corresponding root. Figure 3 depicts Newton's method for κ; to render the iterates themselves, and not merely their limits, we color each point z_0 by blending the colors of the three roots with relative weights $|z_n - r|^{-3}$, for $r \in \{r_0, r_+, r_-\}$, where $z_n := N_\kappa^n(z_0)$. This coloring is a continuous function of z_0, blurred for small n; as n increases, it sharpens into large regions of nearly constant color, separated by narrow regions where the color varies rapidly, that is, where the coloring is nearly discontinuous.

[7] The point ∞ is a repelling fixed point of N_ρ: if $|z|$ is large, then $N_\rho(z) = \frac{2}{3}z + O(1)$. Hence ∞ lies on the Newton fractal of ρ, and $f_0(\infty) = \infty = f_1(\infty)$.

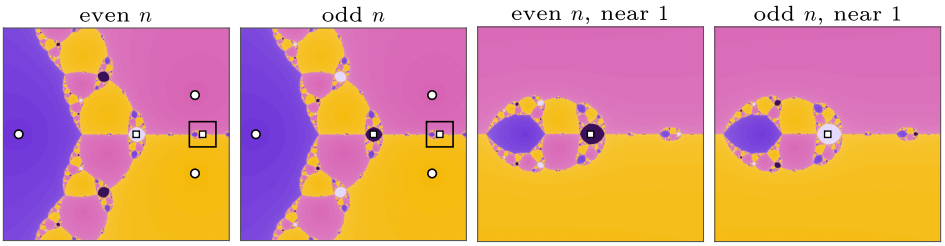


FIGURE 2. The deep computations of Newton's method for $\rho(z) = z^3 - 2z + 2$: the limits of the even iterates N_ρ^{2k} and of the odd iterates N_ρ^{2k+1}, on the whole plane and near the point 1. Points in the basins of the real root and of the roots in the upper and lower half planes (white circles) are purple, orchid and gold, paler where convergence is slower. The remaining open region is the basin of the superattracting cycle $0 \mapsto 1 \mapsto 0$ (white squares). A point there is lilac if the iterate in question is near 0 and plum if it is near 1, so the two colors exchange between even and odd iterates: the two deep computations agree on the basins of the roots and differ on the basin of the cycle. The boundaries of the colored regions form the Newton fractal of ρ.

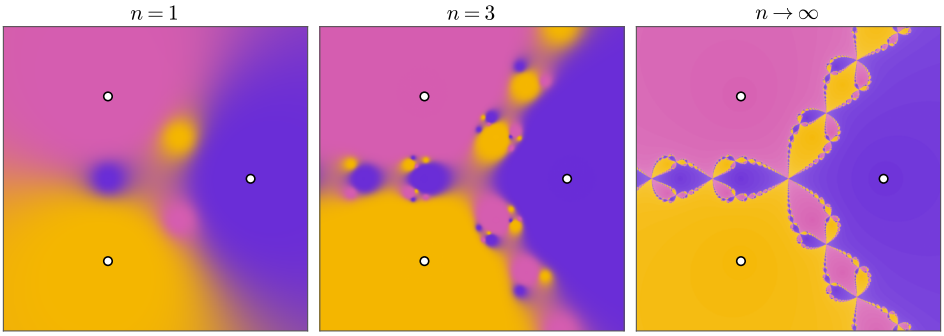


FIGURE 3. Newton's method for $\kappa(z) = z^3 - 1$ after one and three iterations, and its limit. The coloring blends the colors of the three cube roots of unity (white circles) according to the distance of $N_\kappa^n(z)$ to each of them; the limit is constant on each basin and discontinuous across the Newton fractal; in the right panel, points are paler where convergence is slower.

The chaos of Newton's method on its Julia set is visible to the predicates P_1, P_2, P_3; in Proposition 5.10 below we show that it can prevent a Newton CCS on the Riemann sphere from being NIP.

4.3. **Example: Square roots.** We return to the first motivating example of Section 2.5, Newton's method on the real projective line for the polynomial $p(x) =$

$x^2 - a$ with real $a > 0$, whose iterates approximate the square roots $\pm\sqrt{a}$, and work out its deep computations, which place it at the level NIP_3 (see Section 5.2). As for every rational function with real coefficients, the Newton map N_p maps the real projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$ into itself, and the Newton CCS for p may be regarded as having state space $\mathbb{RP}^1$.

Let $\mathbb{R} \to S^1$, $x \mapsto (P_1(x), P_2(x))$, be the inverse stereographic projection of the real line onto $S^1 \subseteq \mathbb{R}^2$, given by

$$P_1(x) = \frac{2x}{x^2+1}, \qquad\qquad P_2(x) = \frac{x^2-1}{x^2+1},$$

and extended to a homeomorphism of the real projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$ onto S^1 by $\infty \mapsto (P_1(\infty), P_2(\infty)) = (0, 1)$; these predicates are the restrictions to $\mathbb{RP}^1$ of the predicates P_1 and P_3 of the Riemann sphere, and they make $(\mathbb{RP}^1, (P_1, P_2))$ a compact computation states structure. The Newton map N_p is

$$x \mapsto x - \frac{x^2-a}{2x} = \begin{cases} \frac{x^2+a}{2x}, & x \neq 0, \infty; \\ \infty, & x = 0, \infty. \end{cases}$$

From every initial point $x_0 \neq 0, \infty$, the Newton steps converge to one of the square roots $\pm\sqrt{a}$; more precisely, the iterates of N_p converge pointwise to the map $\mathfrak{f}\colon \mathbb{RP}^1 \to \mathbb{RP}^1$ given by

$$\mathfrak{f}(x) = \lim_{n\to\infty} N_p^n(x) = \begin{cases} \sqrt{a}, & x \in (0, \infty); \\ -\sqrt{a}, & x \in (-\infty, 0); \\ \infty, & x = 0, \infty. \end{cases}$$

The map $\mathfrak{f}$ is therefore discontinuous at 0 and at ∞ (see Figure 1).

5. Computations across the hierarchy

Every level of the hierarchy

$$\text{stable} \Rightarrow \mathrm{NIP}_3 \Rightarrow \mathrm{NIP}_2 \Rightarrow \mathrm{NIP}_1 \Rightarrow \mathrm{NIP}$$

of Section 3, though defined through the topology of the space of deep computations, is attained by classical algorithms, and some familiar algorithms lie beyond the hierarchy; in this section we establish both facts, and with them the strictness of every implication. In each example the computations are compositions of the steps of an algorithm. At all levels but one they are the powers of a single step map; at the level NIP but not NIP_1 they form a group generated by two steps, both in the Euclidean algorithm and in free reduction in the free group on two generators. The level of each algorithm is then a theorem about the closure of its computations, and for such computations the hierarchy coincides with a hierarchy studied in topological dynamics by Glasner and Megrelishvili [GM22]. We begin with this coincidence (see Proposition 5.1); Table 2 at the end of the section summarizes the examples.

5.1. **Iterative algorithms and tame dynamics.** The algorithms of this section act by continuous steps on a compact space of states, and within this setting, which we now isolate, our hierarchy coincides with the hierarchy of tame dynamical systems of Glasner and Megrelishvili; the coincidence allows their results to be brought to bear on the examples that follow. A *dynamical CCS* is a compositional computation structure $(L, \mathcal{P}, \Gamma)$ in which L is a compact metrizable space, $\mathcal{P}$ is a

countable family of continuous predicates that separates the points of L, and Γ is a countable semigroup of continuous maps $L \to L$ that contains the identity. It is a *cascade CCS* if $\Gamma = \{\gamma^n : n \in \mathbb{N}\}$ is the semigroup of iterates of a single continuous map $\gamma\colon L \to L$. Since $\mathcal{P}$ separates points and L is compact, the map $v \mapsto \mathrm{tp}(v)$ is a homeomorphism of L onto the space of types; hence L is covered by a single shard, every type is realized, and the Extendibility Axiom holds with $\tilde{\gamma} = \gamma$ for every $\gamma \in \Gamma$. The deep computations of a dynamical CCS form the pointwise closure

$$E(L,\Gamma) = \overline{\Gamma} \subseteq L^L,$$

which is the *enveloping semigroup*, introduced by Ellis, of the dynamical system (L,Γ); for a cascade CCS we write $E(L,\gamma)$. The system (L,Γ) is called *tame* if, for every continuous function $f\colon L \to \mathbb{R}$, the family $\{f \circ \gamma : \gamma \in \Gamma\}$ has the NIP of Definition 1.5 [Köh95, Gla06]. We say that a dynamical CCS is *stable*, NIP, NIP_1, NIP_2 or NIP_3 when its set Γ of computations has the corresponding property of Definition 2.8, of condition (ii) of Theorem 3.1 or of Definition 3.2, the last two with respect to a single sizer $r_\bullet$ with $L[r_\bullet] = L$.

A map $p\colon L \to L$ is *of Baire class* 1 if $f \circ p$ is of Baire class 1 for every continuous $f\colon L \to \mathbb{R}$. It suffices that $P \circ p$ be of Baire class 1 for every $P \in \mathcal{P}$. Indeed, the type map embeds L into $\mathbb{R}^{\mathcal{P}}$, and $\mathcal{P}$ is countable, so every open subset of L is a countable union of finite intersections of sets $P^{-1}(U)$, with $P \in \mathcal{P}$ and $U \subseteq \mathbb{R}$ open; by Fact 1.2, the preimage of every open set under p is then an F_σ set, and so is the preimage of every open set under $f \circ p$, which is therefore of Baire class 1, again by Fact 1.2. In general, this definition is less restrictive than the one adopted in Section 1, according to which a map from L to L is of Baire class 1 when it is the pointwise limit of a sequence of continuous maps from L to L (for example, if $L = [0,1] \cup \{2\} \subseteq \mathbb{R}$, the map that sends $[0,1/2]$ to 0 and the rest of L to 2 is of Baire class 1 in the sense just defined, but it is not the pointwise limit of a sequence of continuous maps from L to L, because each such map sends the connected set $[0,1]$ either into $[0,1]$ or to the point 2). For the deep computations of a dynamical CCS, however, the two definitions agree: if every deep computation is of Baire class 1 in the sense just defined, then the CCS is NIP, as we show in Proposition 5.1 below, and the last assertion of Theorem 3.1, applied to the single sizer $r_\bullet$ with $L = L[r_\bullet]$, shows that every deep computation is the pointwise limit of a sequence of computations, which are continuous maps from L to L.

Proposition 5.1. *A dynamical CCS is NIP if and only if the dynamical system (L,Γ) is tame, and if and only if every deep computation is of Baire class* 1*. In that case it is NIP_1, NIP_2 or NIP_3 if and only if $E(L,\Gamma)$ is, respectively, first countable, hereditarily separable or metrizable.*

Proof. The single sizer $r_\bullet$ with $L = L[r_\bullet]$ forms an exhaustive collection that confines Γ. If (L,Γ) is tame, the CCS is NIP: take $f = P$ for $P \in \mathcal{P}$. If the CCS is NIP, then, by Theorem 3.1 and Proposition 1.8, the function $P \circ p$ is of Baire class 1 for every $p \in E(L,\Gamma)$ and $P \in \mathcal{P}$, so every deep computation is of Baire class 1. Finally, suppose that every deep computation is of Baire class 1, and let $f\colon L \to \mathbb{R}$ be continuous. The map $p \mapsto f \circ p$ is continuous on L^L, so the closure of $\{f \circ \gamma : \gamma \in \Gamma\}$ is the compact set $\{f \circ p : p \in E(L,\Gamma)\}$, which lies in $\mathrm{B}_1(L)$. By Theorem 1.6, the family $\{f \circ \gamma : \gamma \in \Gamma\}$ has the NIP, so (L,Γ) is tame. The last statement is Definition 3.2, since $\overline{\tilde{\Gamma}|_L} = E(L,\Gamma)$. □

For metrizable systems, Glasner and Megrelishvili call the tame systems whose enveloping semigroup is first countable, hereditarily separable or metrizable *tame*$_1$, *tame*$_2$ and *hereditarily non-sensitive*, respectively. They prove the inclusions between these classes [GM22, Proposition 0.6] and show by examples that the inclusions are proper. By Proposition 5.1, a dynamical CCS is NIP$_1$, NIP$_2$ or NIP$_3$ exactly when its dynamical system is tame$_1$, tame$_2$ or hereditarily non-sensitive, respectively.

The simplest iterative methods are those whose iterates converge pointwise. In Proposition 5.2 we show that all of them are NIP$_3$, and that the continuity of the limit decides whether they are stable.

Proposition 5.2. *Let $(L, \mathcal{P}, \Gamma)$ be a cascade CCS, with $\Gamma = \{\gamma^n : n \in \mathbb{N}\}$ for a continuous map $\gamma\colon L \to L$, and suppose that the iterates γ^n converge pointwise to a map $\gamma_\infty\colon L \to L$. Then $E(L, \gamma) = \{\gamma^n : n \in \mathbb{N}\} \cup \{\gamma_\infty\}$, and the CCS is NIP$_3$. It is stable if and only if γ_∞ is continuous.*

Proof. A convergent sequence, together with its limit, is a compact subset of the Hausdorff space L^L, hence closed; so $E(L, \gamma)$ consists of the iterates and γ_∞. For each $P \in \mathcal{P}$, the map $P \circ \gamma_\infty$ is the pointwise limit of the continuous functions $P \circ \gamma^n$, so it is of Baire class 1, and, by Proposition 5.1, the CCS is NIP. The space $E(L, \gamma)$ is a countable compact Hausdorff space, hence metrizable, so the CCS is NIP$_3$. Its members other than γ_∞ are continuous, which gives the statement about stability. □

5.2. **Stable and metrizable.** Our descent through the hierarchy starts at its top two levels, stable and NIP$_3$, at which the deep computations form a metrizable space; there we place a rotation of the circle, Newton's method for square roots, the power method and the logistic map for $0 < r \leq 3$. By Proposition 5.2, every convergent method lies at one of these two levels; convergence, however, is not necessary for stability, as the first example shows.

A stable example: rotation. Let $L = \mathbb{T} = \mathbb{R}/\mathbb{Z}$, with the predicates $\cos 2\pi x$ and $\sin 2\pi x$, and let $\gamma(x) = x + \alpha$, where α is irrational. A pointwise limit of rotations is a rotation, and the points $n\alpha$, $n \in \mathbb{N}$, are dense in $\mathbb{T}$; hence the deep computations are all the rotations $x \mapsto x + c$, $c \in \mathbb{T}$. They are continuous, so the rotation CCS is stable, and its space of deep computations is the circle group, a metrizable compactum.

NIP$_3$ but not stable: square roots. The iterates of the Newton map N_p of $p(x) = x^2 - a$, with $a > 0$, on the real projective line (see Section 4.3) converge pointwise to the map $\mathfrak{f}$ defined there, which is discontinuous at 0 and ∞. By Proposition 5.2, the square-root CCS is NIP$_3$ but not stable.

NIP$_3$ but not stable: the power method. The power method computes a dominant eigenvector of a matrix by repeated multiplication and normalization. Let $d \geq 2$ be an integer, let $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_d)$, where $\lambda_1 > \lambda_2 > \cdots > \lambda_d > 0$, let $L = \mathbb{RP}^{d-1}$ be the real projective space of lines through the origin in $\mathbb{R}^d$, in which $[v]$ denotes the line spanned by a nonzero vector v, and let $\mathcal{P}$ consist of the predicates $p_{ij}([v]) = v_i v_j / \|v\|^2$, for $1 \leq i, j \leq d$, the entries of the orthogonal projection onto the line $[v]$. The step of the algorithm is $\gamma([v]) = [Av]$. For $[v] \in \mathbb{RP}^{d-1}$, let $i(v)$ be the least index i with $v_i \neq 0$, and put $\pi([v]) = [e_{i(v)}]$, where $e_1, \ldots, e_d$ is the standard basis of $\mathbb{R}^d$.

Proposition 5.3. *The deep computations of the power method are* $E(L,\gamma) = \{\gamma^n : n \in \mathbb{N}\} \cup \{\pi\}$. *Hence the power method is NIP*$_3$. *Its deep computation* π, *the only one that is not an iterate, is of Baire class* 1 *and discontinuous, so the power method is not stable.*

Proof. For $i = i(v)$, we have $\lambda_i^{-n} A^n v = v_i e_i + \sum_{j>i} (\lambda_j/\lambda_i)^n v_j e_j \to v_i e_i$, so $\gamma^n \to \pi$ pointwise, and Proposition 5.2 applies. The map π is discontinuous at every $[v]$ with $i(v) > 1$, since $[v + te_1] \to [v]$ as $t \to 0$ while $\pi([v + te_1]) = [e_1] \neq \pi([v])$. □

The deep computation π, which sends each input to the eigenline of the largest eigenvalue whose eigenvector has a nonzero component in the input, is not computable in the sense of Section 2, since it is discontinuous on the space of types L; yet it is the limit of a sequence of computations, and the closure in which it lives is countable and metrizable.

Both levels: the logistic map. The logistic map $f_r(x) = rx(1-x)$, with $0 < r \leq 4$, maps $[0,1]$ into itself. Its CCS has states $L = [0,1]$, the single predicate $x \mapsto x$, and the semigroup of iterates of f_r. For $0 < r \leq 1$, let ℓ_r be the constant map 0; for $1 < r \leq 3$, let $\ell_r(x) = 1 - 1/r$ for $0 < x < 1$, and $\ell_r(0) = \ell_r(1) = 0$.

Proposition 5.4. *For* $0 < r \leq 3$, *the iterates of* f_r *converge pointwise to* ℓ_r. *Consequently the logistic CCS is stable for* $0 < r \leq 1$, *and NIP*$_3$ *but not stable for* $1 < r \leq 3$.

Proof. Write $f = f_r$ and $p = 1 - 1/r$, and note that $f(x)/x = r(1-x)$ for $x > 0$. If $r \leq 1$, then $f(x) < x$ for $0 < x \leq 1$, so every orbit decreases to a fixed point of f, and 0 is the only fixed point in $[0,1]$.

Suppose that $1 < r \leq 2$. Then $0 < p \leq \frac{1}{2}$, and $f(x) > x$ for $0 < x < p$, while $f(x) < x$ for $p < x \leq 1$. For $0 < x < 1$ we have $f(x) \in (0, \frac{r}{4}] \subseteq (0, \frac{1}{2}]$, and f is increasing on $[0, \frac{1}{2}]$ and fixes p. Hence the orbit of $f(x)$ is increasing and bounded by p if $f(x) < p$, and decreasing and bounded below by p if $f(x) > p$. In both cases, and trivially if $f(x) = p$, it converges to a fixed point of f in $(0, \frac{1}{2}]$, that is, to p.

Suppose that $2 < r \leq 3$, and put $c = \frac{r}{4}$, the maximum of f, and $b = f(c)$. We have $p < c$, since $c - p = (r-2)^2/(4r)$, and therefore $b < f(p) = p$, since f is strictly decreasing on $[\frac{1}{2}, 1]$. We also have $b \geq 1/r$, because $b - 1/r = (r^3(4-r) - 16)/(16r)$ and $r^3(4-r)$ increases from 16 on $[2,3]$. The concave function f takes the value p at $1/r$ and at p, so $f \geq p$ on $[1/r, p]$. Hence f maps $J_- = [b,p]$ into $J_+ = [p,c]$, and, being decreasing on J_+, it maps J_+ onto J_-. Let $0 < x < 1$. The orbit of $f(x)$ lies in $(0,c]$, and it meets $J_- \cup J_+ = [b,c]$: otherwise all its points would lie in $(0,b)$, where $f(y) > y$ because $b < p$, and the orbit would increase to a fixed point of f in $(0,b]$, which is impossible, since the only positive fixed point of f is p and $p > b$. Since f maps $J_- \cup J_+$ into itself, it remains to be shown that the orbit of every $y \in J_- \cup J_+$ converges to p. The points of period 2 of f that are not fixed are the roots of $r^2y^2 - r(r+1)y + (r+1)$, whose discriminant $r^2(r+1)(r-3)$ is negative for $r < 3$; for $r = 3$ the only root is p. Hence the function $f(f(y)) - y$ vanishes on $[0,1]$ only at 0 and p, and it is negative on $(p,1]$, since it is negative at 1. For $y \in J_+$ with $y > p$, the point $f(f(y))$ lies in J_+, so $p \leq f(f(y)) < y$. The sequence $(f^{2n}(y))$ therefore decreases to a fixed point of $f \circ f$ in $[p,y]$, that is, to p, and $f^{2n+1}(y) \to f(p) = p$. Every $y \in J_-$ satisfies $f(y) \in J_+$, so its orbit converges to p as well.

Finally, $f(0) = f(1) = 0$. In all cases the iterates converge pointwise to ℓ_r. The map ℓ_r is continuous for $r \leq 1$, and discontinuous at 0 for $r > 1$, so Proposition 5.2 gives the remaining statements. □

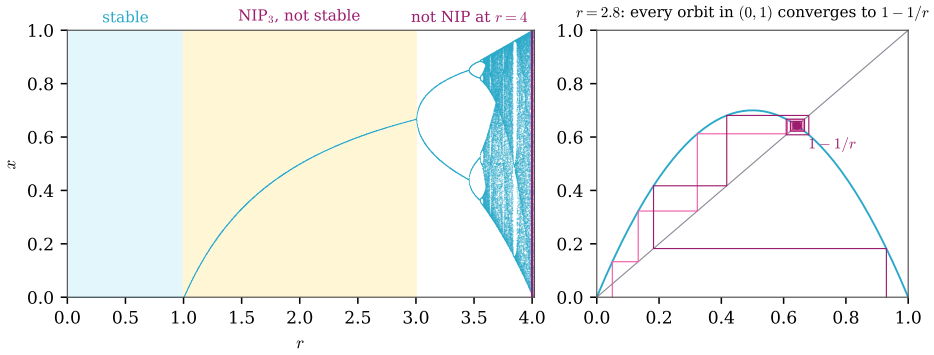


FIGURE 4. The logistic family $f_r(x) = rx(1-x)$. Left: the long-run values of the orbit of 0.3 for each r; the CCS is stable for $r \leq 1$, NIP_3 but not stable for $1 < r \leq 3$ (see Proposition 5.4), and not NIP at $r = 4$ (see Proposition 5.11); the parameters $3 < r < 4$ are not classified here. Right: for $r = 2.8$, two orbits converge to the fixed point $1 - 1/r$; the deep computation sends 0 and 1 to 0 and every other state to $1 - 1/r$.

We return to the logistic family at $r = 4$, and to maps of the interval in general, at the end of the section.

5.3. **Hereditarily separable but not metrizable.** Bresenham's line algorithm in the plane, which we analyze in this subsection, lies at the level NIP_2 but not NIP_3, immediately below the metrizable level. Below that level, no countable set of tests, that is, of pairs of an input and a feature (see Section 3.2), separates the deep computations (see Proposition 3.4). Rounding gives rise to spaces of deep computations of this kind, since an algorithm that rounds at each step takes a new discrete decision at each step.

Drawing a line. Fix an irrational number α with $0 < \alpha < 1$. To draw the line $y = \alpha x + \rho$ on a screen, Bresenham's algorithm [Bre65] moves one pixel to the right at each step and decides whether to move one pixel up. The n-th decision is $b_n(\rho) = \lfloor (n+1)\alpha + \rho \rfloor - \lfloor n\alpha + \rho \rfloor = [\![\{n\alpha + \rho\} \in [1-\alpha, 1)]\!]$, where $\{\cdot\}$ denotes the fractional part, and the state of the algorithm is the error term ρ modulo 1. The step $\rho \mapsto \rho + \alpha$ is a rotation of the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, but the decision bit is not a continuous function of ρ. The standard remedy [MH40, Lot02] is to take the decision sequences themselves as states.

Let $D = \{-k\alpha : k \in \mathbb{Z}\} \subseteq \mathbb{T}$, a countable dense set containing both endpoints 0 and $1 - \alpha$ of the decision interval. The *split circle* $\mathbb{T}_D$ is obtained from $\mathbb{T}$ by replacing each $t \in D$ by two points $t^- < t^+$, with the circular order topology; it is a compact metrizable space, homeomorphic to the Cantor set. The rotation by α preserves D and lifts to a homeomorphism R of $\mathbb{T}_D$ that preserves the labels $\pm$. The decision interval becomes clopen in $\mathbb{T}_D$, and the coding map $\mathbb{T}_D \to 2^{\mathbb{Z}}$,

$y \mapsto ([\![R^n y \in [1-\alpha, 1)]\!])_{n\in\mathbb{Z}}$, is a homeomorphism onto the *Sturmian shift* X_α that conjugates R to the shift σ. The *Bresenham CCS* has states $L = X_\alpha$, the coordinate predicates $P_j(y) = y_j$ for $j \in \mathbb{Z}$, and the semigroup $\Gamma = \{\sigma^n : n \in \mathbb{N}\}$.

A point of $\mathbb{T}_D$ lies *over* $y_0 \in \mathbb{T}$ if the natural projection $\mathbb{T}_D \to \mathbb{T}$ maps it to y_0. For $c \in \mathbb{T}$, let R_c^+ (respectively R_c^-) be the map of $\mathbb{T}_D$ that sends a point over y_0 to $(y_0+c)^+$ (respectively $(y_0+c)^-$) when $y_0+c \in D$, and to y_0+c otherwise; through the coding map, we regard R_c^+ and R_c^- as maps of X_α. The *double arrow circle* $\mathbb{D}$ is $\mathbb{T} \times \{-,+\}$ with the lexicographic circular order and its order topology, and we also write c^- and c^+ for its points $(c,-)$ and $(c,+)$; it is compact, first countable, hereditarily separable and not metrizable [Eng89, 3.10.C].

Theorem 5.5. *The Bresenham CCS is NIP. Its deep computations are*

$$E(X_\alpha, \sigma) = \{\sigma^n : n \in \mathbb{N}\} \cup \{R_c^+, R_c^- : c \in \mathbb{T}\},$$

where each σ^n is isolated and $c^\pm \mapsto R_c^\pm$ is a homeomorphism of $\mathbb{D}$ onto the remaining part. In particular, the Bresenham CCS is NIP$_2$ but not NIP$_3$.

Proof. For NIP, note that the predicates $P_j \circ \sigma^n$ are $\{0,1\}$-valued, and that the j-th coordinate of $\sigma^n y$ is 1 exactly when $R^{n+j}y$ lies in the clopen arc $[1-\alpha, 1)$ of $\mathbb{T}_D$, that is, when y lies in an arc of $\mathbb{T}_D$. A family of arcs of a circularly ordered set has VC dimension at most 3, so, by the remark at the end of Section 1.5, the characteristic functions of its members have the NIP.

For the deep computations, let γ be the limit of σ^{n_i} along an ultrafilter, and let $c = \lim n_i\alpha$ in $\mathbb{T}$. If n_i is eventually constant, then γ is a power of σ. Otherwise, since the points $n\alpha$ are distinct, the ultrafilter decides whether $n_i\alpha$ approaches c from the right or from the left. In the first case, for every point y over y_0 the points $R^{n_i}y$ over $y_0 + n_i\alpha$ approach $y_0 + c$ strictly from the right, and they converge in $\mathbb{T}_D$ to $(y_0+c)^+$ if $y_0 + c \in D$ and to $y_0 + c$ otherwise; so $\gamma = R_c^+$. The second case gives R_c^-. Conversely, every c is approached from both sides by the points $n\alpha$ with $n \geq N$, for every N, so every $R_c^\pm$ is a deep computation, and $R_c^+ \neq R_c^-$, since they differ at the points over $d - c$ with $d \in D$. The same argument shows that $c_i^{\varepsilon_i} \to c^+$ in $\mathbb{D}$ implies $R_{c_i}^{\varepsilon_i} \to R_c^+$ pointwise, and symmetrically for c^-; so $c^\pm \mapsto R_c^\pm$ is a continuous bijection from the compact space $\mathbb{D}$ onto a Hausdorff space, hence a homeomorphism. Finally, fix $d \in D$ and consider the two points $(d - n\alpha)^\pm$ of $\mathbb{T}_D$, which σ^n sends to d^- and d^+. Every map $R_c^\pm$ sends them to the same point, and so does every pointwise limit of σ^{m_i} with $m_i \to \infty$, by the argument above. Hence no net of such maps, and no net of maps σ^m with $m \neq n$ and m bounded, converges to σ^n, so σ^n is isolated. Thus $E(X_\alpha, \sigma)$ is the union of a countable set and a copy of $\mathbb{D}$; it is hereditarily separable and not metrizable. □

The two deep computations R_c^+ and R_c^- are the two ways of drawing the line infinitely far away, where the accumulated error has reached c: they agree except at the pixels where the line passes exactly through a corner, and there they apply the two opposite rounding conventions. Figure 5 shows the two conventions.

The space $E(X_\alpha, \sigma)$ is the closure of a countable set, so it is a separable Rosenthal compactum; it is hereditarily separable and not metrizable, so, by Todorčević's trichotomy (see Theorem 3.3), it contains a copy of the split Cantor set $S(2^{\mathbb{N}})$. For the action of $\mathbb{Z}$ generated by the shift, Glasner and Megrelishvili proved that Sturmian systems are tame$_2$ but not hereditarily non-sensitive [GM22, Corollary 8.3].

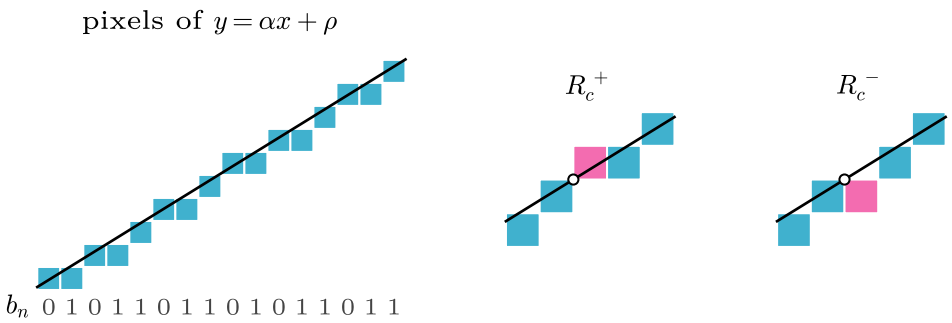


FIGURE 5. Left: the pixels drawn by Bresenham's algorithm for $\alpha = (\sqrt{5}-1)/2$, with the decision bits b_n. Right: where the line passes exactly through a pixel corner, the deep computations R_c^+ and R_c^- break the tie in opposite directions; they agree on every other pixel.

5.4. **First countable but not hereditarily separable.** When two rounding algorithms run in parallel, the computation descends a further level, as we show here in the case of Bresenham's algorithm for lines in three-dimensional space, which lies at the level NIP_1 but not NIP_2. At this level each deep computation is still separated from all the others by countably many tests of its own, but some finitely separated set of deep computations has the cardinality of the continuum (see Proposition 3.4).

Drawing a line in space. Let α and β be irrational numbers in $(0,1)$ such that $1, \alpha, \beta$ are linearly independent over $\mathbb{Q}$. To draw a line in $\mathbb{Z}^3$ with direction $(1, \alpha, \beta)$, the three-dimensional Bresenham algorithm runs the planar algorithm in two coordinates at once. Its CCS has states $L = X_\alpha \times X_\beta$, the coordinate predicates of both factors, and the semigroup generated by $\sigma \times \sigma$.

Theorem 5.6. *The three-dimensional Bresenham CCS is NIP_1 but not NIP_2.*

Proof. Each predicate is a coordinate of one factor, so the CCS is NIP by Theorem 5.5. The deep computations form a closed subspace of $E(X_\alpha, \sigma) \times E(X_\beta, \sigma)$, which is first countable as a product of two first countable spaces; hence the CCS is NIP_1. By Kronecker's theorem, the points $(n\alpha, n\beta)$ with $n \geq N$ are dense in $\mathbb{T}^2$ for every N, so every point (c, c') is approached by them from inside each of the four open quadrants at (c, c'); as in the proof of Theorem 5.5, it follows that every pair $(R_c^\varepsilon, R_{c'}^{\varepsilon'})$, with $\varepsilon, \varepsilon' \in \{+, -\}$, is a deep computation. Consider the pairs (R_c^+, R_{-c}^+), for $c \in \mathbb{T}$. For $0 < \delta < \frac{1}{2}$, the set $[c^+, (c+\delta)^-) \times [(-c)^+, (-c+\delta)^-)$ is a neighborhood of $(c^+, (-c)^+)$ in $\mathbb{D} \times \mathbb{D}$, and it contains $(s^+, (-s)^+)$ only if $c \leq s < c + \delta$ and $-c \leq -s < -c + \delta$, that is, only if $s = c$. These pairs therefore form an uncountable discrete subspace, and the space of deep computations is not hereditarily separable. □

The uncountable discrete set consists of the deep computations that break ties upward in both coordinates at points where the two accumulated errors are opposite. By Todorčević's trichotomy, the space of deep computations contains a copy of $D(2^{\mathbb{N}})$ or of $\hat{D}(S(2^{\mathbb{N}}))$. Each of the two planar algorithms is NIP_2, and drawing in two directions at once leaves the class, just as the Sorgenfrey plane shows that

a product of two hereditarily separable spaces need not be hereditarily separable; Glasner and Megrelishvili show similarly that a product of two tame$_2$ systems need not be tame$_2$ [GM22, Proposition 4.1].

5.5. **NIP but not first countable: the Euclidean algorithm.** The last tame level, NIP but not NIP$_1$, is reached by the Euclidean algorithm, acting on the real projective line through the group generated by its steps. At this level some deep computation is not determined by countably many tests of its own (see Proposition 3.4). The test that always fails has this property whenever uncountably many exact equality tests are also deep computations, for a countable set of inputs then misses one of the tested points, say η, and is therefore unable to distinguish the test that always fails from the equality test at η. The Euclidean algorithm gives rise to equality tests of precisely this kind among its deep computations.

Let $L = \mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$, with the predicates P_1, P_2 of Section 4.3, and let $\Gamma = \mathrm{PGL}(2, \mathbb{Z})$ be the group of Möbius maps $x \mapsto (ax + b)/(cx + d)$ with integer coefficients and $ad - bc = \pm 1$. It is generated by the steps $x \mapsto x + 1$ and $x \mapsto 1/x$ of the Euclidean algorithm, and the maps that the continued fraction algorithm composes are elements of Γ. We call $(\mathbb{RP}^1, (P_1, P_2), \Gamma)$ the *Euclidean CCS*. It is a dynamical CCS whose semigroup is a group, and its deep computations form $E(\mathbb{RP}^1, \Gamma)$.

For an irrational number $\eta = [a_0; a_1, a_2, \dots]$, let p_k/q_k be its convergents, put $p_{-1} = 1$ and $q_{-1} = 0$, and let $\eta_{k+1} = [a_{k+1}; a_{k+2}, \dots]$ be its complete quotients. Let M_k be the Möbius map $t \mapsto (p_k t + p_{k-1})/(q_k t + q_{k-1})$, which belongs to Γ because $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k+1}$. Writing $A_j(t) = a_j + 1/t$, we have $M_k = A_0 \circ A_1 \circ \cdots \circ A_k$, so $M_k^{-1} = A_k^{-1} \circ \cdots \circ A_0^{-1}$ with $A_j^{-1}(t) = 1/(t - a_j)$. Thus M_k^{-1} is the map that the first $k + 1$ steps of the continued fraction algorithm for η apply, and the classical identity $\eta = M_k(\eta_{k+1})$ says that it sends η to its complete quotient η_{k+1}.

Lemma 5.7. *Let η be an irrational number, and let $x \in \mathbb{RP}^1$ be distinct from η. Then $M_k^{-1}(x) + q_{k-1}/q_k \to 0$ as $k \to \infty$, while $M_k^{-1}(\eta) = \eta_{k+1} > 1$.*

Proof. Solving $M_k(t) = x$ gives $M_k^{-1}(\infty) = -q_{k-1}/q_k$ and, for real x,

$$M_k^{-1}(x) = \frac{q_{k-1}x - p_{k-1}}{p_k - q_k x} = -\frac{q_{k-1}}{q_k} \cdot \frac{x - p_{k-1}/q_{k-1}}{x - p_k/q_k}$$

for $k \geq 1$ (for such k we have $q_{k-1} \geq 1$), where the value is ∞ if $x = p_k/q_k$. The convergents of an irrational number are pairwise distinct, so x equals p_k/q_k for at most one k; for all larger k the formula gives a real number. Both convergents tend to $\eta \neq x$, so the last fraction tends to 1, and $0 < q_{k-1}/q_k \leq 1$ for $k \geq 1$; hence $M_k^{-1}(x) + q_{k-1}/q_k \to 0$. The identity $M_k^{-1}(\eta) = \eta_{k+1}$ is the classical one, and $\eta_{k+1} > a_{k+1} \geq 1$ because η_{k+1} is irrational. □

Let $\varphi = (1 + \sqrt{5})/2$. Let S be the set of irrational numbers $\eta = [0; a_1, a_2, \dots]$ whose sequence of partial quotients is a concatenation $1^1 d_1 1^2 d_2 1^3 d_3 \cdots$, where 1^n denotes n consecutive ones and each $d_n \in \{1, 2\}$. The block 1^n occupies the indices $s_n, \dots, s_n + n - 1$ with $s_n = n(n+1)/2$. Distinct choices of (d_n) give distinct numbers, by the uniqueness of continued fraction expansions, so S has the cardinality of the continuum.

Proposition 5.8. *(i) For every $\eta \in S$, the map δ_η that sends η to φ and every other point of $\mathbb{RP}^1$ to $-1/\varphi$ is a deep computation of the Euclidean CCS.*
(ii) The constant map c with value $-1/\varphi$ is a deep computation, and it has no countable base of neighborhoods in $E(\mathbb{RP}^1, \Gamma)$.
Consequently $E(\mathbb{RP}^1, \Gamma)$ is not first countable.

Proof. (i) Let $\eta \in S$, and for each n let $k_n = s_n + \lfloor n/2 \rfloor$ be the middle index of the block 1^n, and $m_n = \lceil n/2 \rceil - 1$. Then $a_{k_n} = a_{k_n - 1} = \cdots = a_{k_n - m_n} = 1$ and $a_{k_n+1} = \cdots = a_{k_n+m_n} = 1$, and $m_n \to \infty$. We have $q_{k-1}/q_k = [0; a_k, a_{k-1}, \ldots, a_1]$ and $\eta_{k+1} = [a_{k+1}; a_{k+2}, \ldots]$. Suppose that two continued fractions, finite or infinite, $[c_0; c_1, \ldots]$ and $[c'_0; c'_1, \ldots]$ satisfy $c_j = c'_j$ for $j < m$, with $c_1 = \cdots = c_{m-1} = 1$. Then they share the convergents of index at most $m - 1$, and both lie between the last two of them (or equal one of them). It follows that they differ by at most $1/(q_{m-2}q_{m-1}) \le 1/(F_{m-1}F_m)$, where F_j are the Fibonacci numbers, whatever the remaining partial quotients are. The hypothesis holds with $m = m_n$ for the pair $[0; a_{k_n}, \ldots, a_1]$, $[0; 1, 1, \ldots]$ and for the pair $[a_{k_n+1}; a_{k_n+2}, \ldots]$, $[1; 1, 1, \ldots]$. Hence $q_{k_n-1}/q_{k_n} \to [0; 1, 1, \ldots] = 1/\varphi$ and $\eta_{k_n+1} \to [1; 1, 1, \ldots] = \varphi$. By Lemma 5.7, the maps $M_{k_n}^{-1} \in \Gamma$ converge pointwise to δ_η, at $x = \infty$ as well.

(ii) Let (η_j) be a sequence of distinct elements of S. For each $x \in \mathbb{RP}^1$, at most one j has $\eta_j = x$, so $\delta_{\eta_j}(x) = -1/\varphi$ for all but at most one j. Hence $\delta_{\eta_j} \to c$ pointwise, and $c \in E(\mathbb{RP}^1, \Gamma)$ because this set is closed. Now let $(U_i)_{i \in \mathbb{N}}$ be open neighborhoods of c in $E(\mathbb{RP}^1, \Gamma)$. Each U_i contains a basic neighborhood $\{f : d(f(x), c(x)) < \varepsilon_i \text{ for } x \in F_i\}$ with $F_i \subseteq \mathbb{RP}^1$ finite, $\varepsilon_i > 0$ and d a metric on $\mathbb{RP}^1$. The set $F = \bigcup_i F_i$ is countable, so some $\eta \in S$ lies outside F, and then δ_η agrees with c on F. Hence $\delta_\eta \in \bigcap_i U_i$, while $\delta_\eta \neq c$. Thus no countable family of neighborhoods of c has intersection $\{c\}$. In a Hausdorff space, the intersection of a base of neighborhoods of c is $\{c\}$, so c has no countable base of neighborhoods. □

Corollary 5.9. *The Euclidean CCS is NIP but not NIP$_1$. Its space of deep computations contains the copy $\{\delta_\eta : \eta \in S\} \cup \{c\}$ of the Alexandroff compactification $A(2^{\mathbb{N}})$.*

Proof. Ellis proved that the action of $\mathrm{GL}(2, \mathbb{R})$ on the projective line is tame [Ell93]; see [GM22, Example 5.7]. For every continuous $f \colon \mathbb{RP}^1 \to \mathbb{R}$, the family $\{f \circ \gamma : \gamma \in \Gamma\}$ is contained in the corresponding family for $\mathrm{GL}(2, \mathbb{R})$, so it has the NIP. Hence the action of Γ is tame, and, by Proposition 5.1, the Euclidean CCS is NIP. By Proposition 5.8, it is not NIP$_1$. Each δ_η is isolated in $\{\delta_\eta : \eta \in S\} \cup \{c\}$, since it is the only member that sends η near φ, and every neighborhood of c contains all but finitely many of the δ_η, as in the proof of Proposition 5.8. The set $\{\delta_\eta : \eta \in S\} \cup \{c\}$ is therefore the one-point compactification of a discrete set of cardinality $2^{\aleph_0}$, and hence it is homeomorphic to $A(2^{\mathbb{N}})$ (see Section 1.4). □

The deep computation δ_η is an exact equality test: it takes one value if the input is exactly η and another value otherwise. It is the pointwise limit of maps that the continued fraction algorithm applies to η, when these maps are applied to all inputs. The mechanism is that of convergence groups: groups of homeomorphisms of a compact space in which every sequence of distinct elements has a subsequence converging to a constant map uniformly on the compact subsets of the complement of some point. By Lemma 5.7, the maps M_k^{-1} draw the images of all points other than η together, near $-q_{k-1}/q_k \in [-1, 0)$, while they send η into $(1, \infty)$; in the

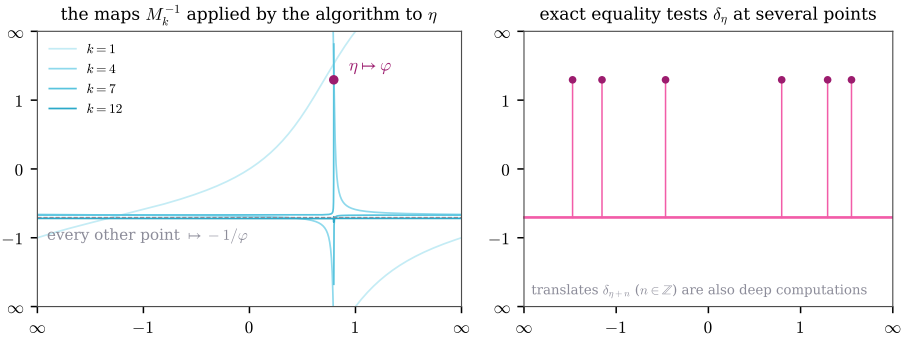


FIGURE 6. The Euclidean algorithm, drawn on the projective line (each axis runs from ∞ through 0 back to ∞). Left: the maps M_k^{-1} that the continued fraction algorithm applies to a number η whose partial quotients contain longer and longer runs of ones; as k grows along the middles of the runs, every point other than η is sent near $-1/\varphi$, while η is sent near φ. Right: the limits, exact equality tests δ_η, at several points. Every neighborhood of the constant map $-1/\varphi$ contains all but finitely many of the uncountably many tests δ_η, so the space of deep computations is not first countable (see Proposition 5.8).

language of convergence groups, every irrational η is thus a conical limit point of Γ, that is, some sequence in Γ sends η to one limit and every other point to another. The choice of S makes the two limits the same for uncountably many η. Glasner and Megrelishvili use the same device to prove that an infinite countable uniform convergence group acting on a compact metric space with infinite limit set is tame but not tame_1 [GM22, Proposition 6.1 and Theorem 6.5]. Their theorem does not cover Γ, which contains parabolic elements and is not a uniform convergence group on $\mathbb{RP}^1$. Ellis's example, the projective action of the whole group $\mathrm{GL}(2,\mathbb{R})$, is also tame but not tame_1 [GM22, Example 5.7]; the Euclidean CCS shows that the countable group generated by the two steps of the algorithm already reaches this level.

Iterating a single element of Γ cannot produce δ_η. For $g \in \Gamma$, the closure of $\{g^n : n \in \mathbb{Z}\}$ in $(\mathbb{RP}^1)^{\mathbb{RP}^1}$ is countable, hence metrizable, and it does not contain δ_η. Indeed, if g has finite order, the set of its powers is finite. If g is parabolic, with fixed point r, then g^n converges pointwise to the constant map r as $n \to \pm\infty$. If g is hyperbolic, with attracting and repelling fixed points a and r, then g^n converges to the map that fixes r and sends every other point to a as $n \to +\infty$, and to the map that fixes a and sends every other point to r as $n \to -\infty$. If g reverses orientation and has infinite order, then its trace t is nonzero, so g^2 has trace $t^2 + 2 > 2$ and is hyperbolic, with the same fixed points as g; the odd powers $g^{2m+1} = g \circ g^{2m}$ have the same two limits as the even ones. In every case the closure consists of the powers of g, which are homeomorphisms, and at most two limit maps, each of which is constant or fixes the unique point at which it differs from a constant. The map δ_η is none of these, since it is not injective and $\delta_\eta(\eta) = \varphi \neq \eta$.

Free reduction. A second example at this level is the free group F_2 on two generators a, b, acting on its boundary ∂F_2, the space of infinite reduced words in a, a^{-1}, b, b^{-1}, which is a Cantor set. A generator acts on an infinite word by prepending a letter and cancelling it against the first letter if they are inverse, as a stack does. With the predicates $w \mapsto [\![w_n = x]\!]$, for $n \in \mathbb{N}$ and each letter x, this is a dynamical CCS, and Glasner and Megrelishvili proved that the system is tame but not tame$_1$ [GM22, Corollary 6.6 and Example 6.7]. By Proposition 5.1, this CCS is NIP but not NIP$_1$.

An open question. Both examples at this level are actions of groups with two generators. Whether a single map can reach this level, that is, whether some cascade CCS is NIP but not NIP$_1$, is Question 7.2. For actions of $\mathbb{Z}$, and more generally of abelian groups, the existence of a tame metrizable system whose enveloping semigroup is not first countable is Question 6.8 of Glasner and Megrelishvili [GM22]; to our knowledge, it is open.

5.6. **Beyond NIP.** Chaotic iteration leads beyond the hierarchy, to sets of computations with the independence property, and we close the section with examples, among them Newton's method for $z^2 - 1$ on the Riemann sphere, the doubling map of the circle and the logistic map at $r = 4$. In a set of computations with the independence property, the computations realize, for some feature and on suitable inputs, every finite pattern of high and low values.

Newton's method on the Riemann sphere. On its Julia set the Newton map is chaotic, and the chaos is visible to the predicates of the CCS. We exhibit the independence in the simplest case, the square roots of 1.

Proposition 5.10. *Let $p(z) = z^2 - 1$, and let $N(z) = \frac{1}{2}\big(z + \frac{1}{z}\big)$ be its Newton map on $\hat{\mathbb{C}}$. The functions $P_2 \circ N^{3k}$, $k \in \mathbb{N}$, where P_2 is the second coordinate of the inverse stereographic projection $\hat{\mathbb{C}} \to S^2$ of Section 4, form an independent sequence: for all finite disjoint sets $E, F \subseteq \mathbb{N}$ there is $z \in \hat{\mathbb{C}}$ with $P_2(N^{3k}z) \geq \frac{1}{2}$ for $k \in E$ and $P_2(N^{3k}z) \leq -\frac{1}{2}$ for $k \in F$. In particular, the Newton CCS of p is not NIP.*

Proof. The Möbius transformation $\psi(z) = (z-1)/(z+1)$ conjugates N to the squaring map: $\psi(N(z)) = \psi(z)^2$. It maps the imaginary axis, together with ∞, onto the unit circle, with $\psi^{-1}(e^{i\theta}) = i\cot(\theta/2)$. Hence, for $z = i\cot(\theta/2)$, we have $N^n(z) = i\cot(2^n\theta/2)$ and

$$P_2\big(N^n(z)\big) = \frac{2\cot(2^{n-1}\theta)}{\cot^2(2^{n-1}\theta) + 1} = \sin(2^n\theta).$$

Write $\theta = 2\pi\sum_{j\geq 1} x_j 2^{-j}$ with bits x_j. The fractional part of $2^{3k}\theta/2\pi$ has binary expansion $0.x_{3k+1}x_{3k+2}x_{3k+3}\cdots$. If $(x_{3k+1}, x_{3k+2}, x_{3k+3}) = (0,1,0)$, this fractional part lies in $[\frac{1}{4}, \frac{3}{8}]$, so $\sin(2^{3k}\theta) \geq \sin\frac{3\pi}{4} > \frac{1}{2}$; if the block is $(1,1,0)$, it lies in $[\frac{3}{4}, \frac{7}{8}]$, so $\sin(2^{3k}\theta) < -\frac{1}{2}$. Choosing the block $(0,1,0)$ for $k \in E$ and $(1,1,0)$ for $k \in F$ gives the required z. □

The proof also shows that the doubling map $x \mapsto 2x$ of the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with the predicates $\cos 2\pi x$ and $\sin 2\pi x$, is not NIP: with $x = \theta/2\pi$, the functions $\sin(2\pi\, 2^{3k}x)$ form an independent sequence. The doubling map shifts the binary digits of x, and the independence comes from the freedom to choose those digits.

On the real projective line, by contrast, the same algorithm is tame. The Julia set of N, the imaginary axis together with ∞, meets $\mathbb{RP}^1$ only at ∞, and on $\mathbb{RP}^1$ the iterates converge to the deep computation of Section 4.3 for $a = 1$; by Proposition 5.2, the corresponding CCS is therefore NIP_3 (see Section 5.2). Whether Newton's method is tame is thus decided by the space of states on which it acts, and not by the formula that defines its step.

The logistic map at $r = 4$. The doubling map also governs the logistic map $f_4(x) = 4x(1-x)$ of Section 5.2.

Proposition 5.11. *The functions f_4^{3k}, $k \in \mathbb{N}$, form an independent sequence on $[0,1]$: for all finite disjoint sets $E, F \subseteq \mathbb{N}$ there is $x \in [0,1]$ with $f_4^{3k}(x) \geq \frac{3}{4}$ for $k \in E$ and $f_4^{3k}(x) \leq \frac{1}{4}$ for $k \in F$. In particular, the logistic CCS for $r = 4$ is not NIP.*

Proof. Let $h(\theta) = \sin^2(\pi\theta)$ for $\theta \in \mathbb{T}$. Then $f_4(h(\theta)) = 4\sin^2(\pi\theta)\cos^2(\pi\theta) = h(2\theta)$, so $f_4^n(h(\theta)) = \sin^2(\pi\, 2^n\theta)$. Write $\theta = \sum_{j\geq 1} x_j 2^{-j}$ in binary. The fractional part u of $2^{3k}\theta$ is then $0.x_{3k+1}x_{3k+2}\cdots$ in binary. If $(x_{3k+1}, x_{3k+2}, x_{3k+3}) = (0,1,1)$, then $u \in [\frac{3}{8}, \frac{1}{2}]$ and $\sin^2(\pi u) \geq \sin^2 \frac{3\pi}{8} > \frac{3}{4}$; if the block is $(0,0,0)$, then $u \in [0, \frac{1}{8}]$ and $\sin^2(\pi u) \leq \sin^2 \frac{\pi}{8} < \frac{1}{4}$. Choosing the block $(0,1,1)$ for $k \in E$ and $(0,0,0)$ for $k \in F$, and putting $x = h(\theta)$, gives the required point. □

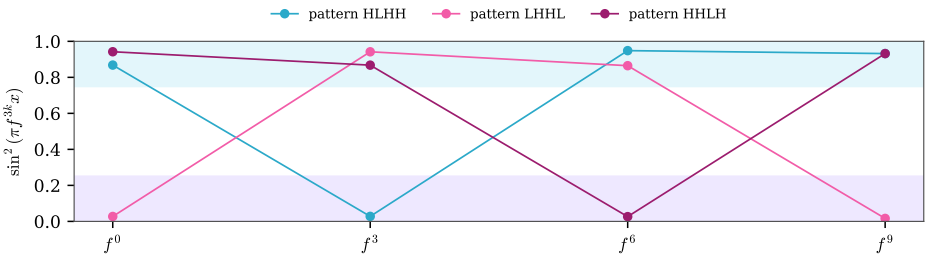


FIGURE 7. Independence for the doubling map. The values of the predicate $\sin^2(\pi x)$ at the times $0, 3, 6, 9$ follow any prescribed pattern of high (H, at least $3/4$) and low (L, at most $1/4$): the initial state is chosen with the block 011 of binary digits for H and 000 for L.

Thus the logistic family, which is stable for $0 < r \leq 1$ and NIP_3 but not stable for $1 < r \leq 3$ (see Proposition 5.4), leaves the hierarchy at $r = 4$. A continuous map of a compact interval is tame if and only if it is not chaotic in the sense of Li and Yorke [Li11, Theorems 4.23 and 4.24], and Glasner and Megrelishvili record, from an unpublished paper, that it is tame if and only if it is hereditarily non-sensitive [GM23, Remark 5.4]. Consequently, for cascade CCSs on $[0,1]$ with the predicate $x \mapsto x$, the levels NIP, NIP_1, NIP_2 and NIP_3 coincide.

The Thue–Morse sequence. Even a sequence produced by a simple substitution rule can be wild. The Thue–Morse sequence t_n is the parity of the number of ones in the binary expansion of n; it is the limit of the words obtained from 0 by the substitution $0 \mapsto 01$, $1 \mapsto 10$. The two-sided sequences all of whose finite blocks

occur in (t_n) form the Thue–Morse subshift $X_{TM} \subseteq 2^{\mathbb{Z}}$, on which the shift σ is a homeomorphism. Glasner and Megrelishvili note that the action of $\mathbb{Z}$ that it generates is not tame [GM23, Example 5.3]. Hence the CCS with states X_{TM}, the coordinate predicates, and the group $\{\sigma^n : n \in \mathbb{Z}\}$ is not NIP, by Proposition 5.1. The cascade CCS of σ is not NIP either. Indeed, the words obtained from 0 by an even number of applications of the substitution are palindromes, and every finite block of (t_n) occurs in one of them, so the reversal ι of two-sided sequences maps X_{TM} onto itself and satisfies $\iota \circ \sigma = \sigma^{-1} \circ \iota$. An independent sequence $f \circ \sigma^{n_k}$, with f continuous and the n_k distinct, has infinitely many indices n_k of the same sign. If they are positive, these terms already form an independent sequence in the cascade; if they are negative, the functions $f \circ \sigma^{n_k} \circ \iota = (f \circ \iota) \circ \sigma^{-n_k}$ form an independent sequence with positive exponents, because ι is a bijection of X_{TM}. By Proposition 5.1, the cascade CCS is not NIP.

5.7. **Summary.** Table 2 collects the examples of this section. The Euclidean algorithm and free reduction are actions of groups with two generators; every other entry is a single map iterated.

Level	Computation	Deep computations; source
stable	rotation of the circle	the circle group
	logistic map, $0 < r \leq 1$	iterates and one continuous limit; Prop. 5.4
NIP_3, not stable	power method	countable; Prop. 5.3
	Newton's method for $x^2 - a$ on $\mathbb{RP}^1$	countable; Prop. 5.2
	logistic map, $1 < r \leq 3$	countable; Prop. 5.4
NIP_2, not NIP_3	Bresenham, plane (Sturmian shift)	isolated iterates and a double arrow circle; Thm. 5.5
NIP_1, not NIP_2	Bresenham, space (two Sturmian shifts)	contains an uncountable discrete set; Thm. 5.6
NIP, not NIP_1	Euclidean algorithm: $\mathrm{PGL}(2,\mathbb{Z})$ on $\mathbb{RP}^1$ (group)	contains $A(2^{\mathbb{N}})$, of exact equality tests; Cor. 5.9
	free reduction: F_2 on ∂F_2 (group)	not first countable; [GM22]
	a single map	open; Question 7.2
not NIP	doubling map of the circle	Prop. 5.10
	Newton's method for $z^2 - 1$ on $\hat{\mathbb{C}}$	Prop. 5.10
	logistic map, $r = 4$	Prop. 5.11
	Thue–Morse shift	[GM23]

TABLE 2. Computations at each level of the hierarchy.

6. Randomized versions of NIP and Monte Carlo computability of deep computations

Replacing topology by measure, we dispense in this section with the countability of the set $\mathcal{P}$ of predicates, on which the theory of Sections 3–5 depends. In those sections we studied deep computations as *topological* limits of computations; under the NIP, they are of the first Baire class on the type shards, and on each shard each of them is the limit of a sequence of computations. That theory rests on the type shards being Polish, and hence on the countability of $\mathcal{P}$, which is used in the proof of Theorem 3.1. Instead of asking that a deep computation be of the first Baire class, we ask that it be *universally measurable*, that is, measurable with respect to every Radon probability measure on the type shards, which allows it to be evaluated almost surely by random sampling. We call such deep computations *universally Monte Carlo computable* (see Definition 6.6). The role of the Bourgain–Fremlin–Talagrand dichotomy is then taken by its measure-theoretic form (see Theorem 6.5), and the role of NIP is shared with *Talagrand stability*, a measure-theoretic condition introduced by Talagrand in his study of Pettis integration (see Definition 6.8).

The main results of the section are Theorem 6.7, which derives Monte Carlo computability from the NIP, and Corollary 6.14, which derives it from universal Talagrand stability.

6.1. **NIP and Monte Carlo computability of deep computations.** Of the two routes to universal Monte Carlo computability followed in this section, the first starts from the NIP and passes through universal measurability, the measure-theoretic counterpart of membership in the first Baire class, which we introduce here. It leads to Theorem 6.7, by which the deep computations of a confined set of computations with the NIP on shards and features are universally Monte Carlo computable. By Fact 1.2, a function $f\colon X \to \mathbb{R}$ on a perfectly normal space X is of Baire class 1 if and only if $f^{-1}[U]$ is an F_σ subset of X for every open $U \subseteq \mathbb{R}$. The following definition replaces the F_σ condition by measurability with respect to every Radon measure.

Definition 6.1. Given a Hausdorff space X and a measurable space (Y, Σ), we say that $f : X \to Y$ is *universally measurable* (with respect to Σ) if $f^{-1}(E)$ is μ-measurable for every Radon measure μ on X and every $E \in \Sigma$. When $Y = \mathbb{R}$, we will always take $\Sigma = \mathcal{B}(\mathbb{R})$, the Borel σ-algebra of $\mathbb{R}$.

If X is a compact Hausdorff space, then Radon measures on X are finite, and every nonzero Radon measure on X becomes a probability measure after normalization. Moreover, the sets $E \subseteq \mathbb{R}$ whose preimages under a given function are μ-measurable form a σ-algebra, which contains the Borel sets as soon as it contains the open sets. Together, these two observations yield the following simplification of Definition 6.1 for compact spaces.

Remark 6.2. If X is compact, then a function $f : X \to \mathbb{R}$ is universally measurable if and only if $f^{-1}(U)$ is μ-measurable for every Radon probability measure μ on X and every open set $U \subseteq \mathbb{R}$.

Universal measurability thus asks for measurability with respect to every Radon measure on the domain, that is, with respect to every admissible way of measuring its subsets; an account of universally measurable sets and functions may be found in the *Handbook of measure theory* [Pap02, Chapters 1 and 2].

Notation 6.3. Following Bourgain, Fremlin and Talagrand [BFT78], we denote by $M_r(X)$ the collection of all universally measurable real-valued functions on X; for a Radon measure μ on X, we denote by $\mathscr{M}^0(X,\mu)$ the collection of all μ-measurable real-valued functions on X.

The maps that arise in the study of deep computations, transitions and deep computations themselves among them, send a state space $L \subseteq \mathbb{R}^{\mathcal{P}}$ into $\mathbb{R}^{\mathcal{P}}$. The product space $\mathbb{R}^{\mathcal{P}}$ carries two natural σ-algebras: the Borel σ-algebra, generated by the open subsets of $\mathbb{R}^{\mathcal{P}}$, and the cylinder σ-algebra, generated by the subbasic open sets $\pi_P^{-1}(U)$ with $P \in \mathcal{P}$ and $U \subseteq \mathbb{R}$ open. In general, the cylinder σ-algebra is strictly smaller, but the two coincide when $\mathcal{P}$ is countable. We use the cylinder σ-algebra to define universally measurable maps into $\mathbb{R}^{\mathcal{P}}$, because of the following characterization.

Proposition 6.4. *Let X be a Hausdorff space and $Y = \prod_{i\in I} Y_i$ be any product of measurable spaces (Y_i, Σ_i) for $i \in I$. Let Σ_Y be the cylinder σ-algebra generated by the measurable spaces (Y_i, Σ_i), and let $\pi_i \colon Y \to Y_i$ be the projections. The following are equivalent for $f : X \to Y$:*

(i) $f : X \to Y$ is universally measurable (with respect to Σ_Y).
(ii) $\pi_i \circ f : X \to Y_i$ is universally measurable (with respect to Σ_i) for all $i \in I$.

Proof. The implication (i)⇒(ii) holds because each projection π_i is measurable and $(\pi_i \circ f)^{-1}(E) = f^{-1}(\pi_i^{-1}(E))$ for every $E \in \Sigma_i$. To prove (ii)⇒(i), fix a Radon measure μ on X, and note that the sets $C \subseteq Y$ for which $f^{-1}(C)$ is μ-measurable form a σ-algebra. This σ-algebra contains every measurable cylinder $C = \prod_{i\in I} C_i$: if J is the finite set of indices $i \in I$ with $C_i \neq Y_i$, then $f^{-1}(C) = \bigcap_{i\in J} (\pi_i \circ f)^{-1}(C_i)$ is μ-measurable by (ii). Since the measurable cylinders generate Σ_Y, the set $f^{-1}(C)$ is μ-measurable for every $C \in \Sigma_Y$. □

By Proposition 6.4, the universal measurability of a map into $\mathbb{R}^{\mathcal{P}}$ can be checked feature by feature, just as membership in the first Baire class can (see Proposition 1.8).

We saw in Section 1 that the NIP corresponds to relative compactness in the space of functions of Baire class 1 (see Theorems 1.4, 1.6 and 1.9), and in Section 3 that, when the set $\mathcal{P}$ of predicates is countable, this correspondence extends to deep computations (see Theorem 3.1). The following theorem, valid on every Hausdorff space, characterizes the NIP in terms of universal measurability; applied to the type shards, it requires no countability of $\mathcal{P}$.

Theorem 6.5 (Bourgain–Fremlin–Talagrand [BFT78, Theorem 2F])**.** *Let X be a Hausdorff space, and let $A \subseteq \mathrm{C_p}(X)$ be pointwise bounded. The following are equivalent:*

(i) $\overline{A} \subseteq M_r(X)$.
(ii) For every compact $K \subseteq X$, $A|_K$ satisfies the NIP.
(iii) For every Radon measure μ on X, A is relatively countably compact in $\mathscr{M}^0(X,\mu)$, i.e., every countable subset of A has a limit point in $\mathscr{M}^0(X,\mu)$.

Theorem 6.5 leads to a measure-theoretic counterpart of computability, the universal Monte Carlo computability of Definition 6.6 below, whose motivation lies in two classical facts.

(1) Littlewood's second principle, that every Lebesgue measurable function is "nearly continuous", is made precise by Luzin's theorem: if (X, Σ, μ) is a finite Radon measure space and Y is a second-countable topological space (e.g., $Y = \mathbb{R}^{\mathcal{P}}$ with $\mathcal{P}$ countable) with its Borel σ-algebra, then a function $f\colon X \to Y$ is measurable if and only if, for every $E \in \Sigma$ and every $\varepsilon > 0$, there is a closed set $F \subseteq E$ such that the restriction $f|_F$ is continuous and $\mu(E \setminus F) < \varepsilon$.
(2) Computability of a transform is characterized by its continuous extendibility to the space of types (see Section 2.3); this characterization is at the core of the first paper of this series [ADIW26].

Taken together, the two facts justify the definition, for if the extension of a transform is universally measurable on a type shard, then, by Luzin's theorem, for every Radon probability measure on the shard each output feature is continuous on a closed subset of the shard whose measure is as close to 1 as desired, and there, by the Stone–Weierstrass theorem, it is a uniform limit of polynomials in the features.

Definition 6.6. Let $(L, \mathcal{P}, \Gamma)$ be a CCS, and let R be a collection of sizers. A transform $f\colon L \to \mathcal{L}_{\mathrm{sh}}$ is *universally Monte Carlo computable on* R if it has an extension $\tilde{f}\colon \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ with the following property: for every $r_\bullet \in R$ there is a sizer $s_\bullet$ such that the restriction $\tilde{f}|_{\mathcal{L}[r_\bullet]}\colon \mathcal{L}[r_\bullet] \to \mathcal{L}[s_\bullet]$ is universally measurable, that is, $\pi_P \circ \tilde{f}|_{\mathcal{L}[r_\bullet]}\colon \mathcal{L}[r_\bullet] \to [-s_P, s_P]$ is μ-measurable for every Radon probability measure μ on $\mathcal{L}[r_\bullet]$ and every $P \in \mathcal{P}$.

The deep computations of a set of computations confined by an exhaustive collection of sizers are transforms $L \to \mathcal{L}_{\mathrm{sh}}$ (see Proposition 2.5). Since each type shard $\mathcal{L}[r_\bullet]$ is compact, Remark 6.2 shows that Radon probability measures suffice in Definition 6.6, in place of general Radon measures.

Theorem 6.7. *Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom, let R be an exhaustive collection of sizers, and let $\Delta \subseteq \Gamma$ be R-confined. If $\pi_P \circ \Delta|_{L[r_\bullet]}$ satisfies the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$, where $\pi_P\colon \mathbb{R}^{\mathcal{P}} \to \mathbb{R}$ is the projection onto the coordinate P, then every deep computation in $\overline{\Delta}$ is universally Monte Carlo computable on R.*

Proof. Let $f \in \overline{\Delta}$, let $\tilde{f}$ be the extension of Lemma 2.7, and fix $P \in \mathcal{P}$ and $r_\bullet \in R$. By the Extendibility Axiom and confinement, $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ is a pointwise bounded set of continuous functions on the compact space $\mathcal{L}[r_\bullet]$. Its restriction to $L[r_\bullet]$ is $\pi_P \circ \Delta|_{L[r_\bullet]}$, which has the NIP by hypothesis, and $L[r_\bullet]$ is dense in $\mathcal{L}[r_\bullet]$; by Lemma 1.10, the set $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ itself has the NIP, and so does its restriction to every compact $K \subseteq \mathcal{L}[r_\bullet]$; Theorem 6.5 then gives $\overline{\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq M_r(\mathcal{L}[r_\bullet])$. By Lemma 2.7, the function $\pi_P \circ \tilde{f}|_{\mathcal{L}[r_\bullet]}$ belongs to this closure, and it is therefore universally measurable. Since $\tilde{f}$ maps $\mathcal{L}[r_\bullet]$ into itself, and since P and $r_\bullet$ are arbitrary, the deep computation f satisfies Definition 6.6 with $s_\bullet = r_\bullet$ for every $r_\bullet \in R$. □

Whether the converse of Theorem 6.7 holds is Question 7.3.

6.2. **Talagrand stability and Monte Carlo computability of deep computations.** The second route to universal Monte Carlo computability starts from a notion closely related to the NIP, which Talagrand introduced in his study of Pettis

integration [Tal84]; we follow this route here, and at the end of the subsection we compare the two. Let X be a compact Hausdorff space, let $A \subseteq \mathbb{R}^X$ be a set of functions, and let μ be a Radon probability measure on X. For a μ-measurable set $E \subseteq X$, a positive integer k and real numbers $a < b$, writing the points of E^{2k} as $x = (x_0, \ldots, x_{2k-1})$, let

$$D_k(A, E, a, b) = \bigcup_{f \in A} \{x \in E^{2k} : f(x_{2i}) \leq a,\ f(x_{2i+1}) \geq b \text{ for all } i < k\}.$$

Definition 6.8. The set A is *Talagrand μ-stable* if, for every μ-measurable set $E \subseteq X$ of positive measure and all reals $a < b$, there is an integer $k \geq 1$ such that

$$(\mu^{2k})^*(D_k(A, E, a, b)) < (\mu(E))^{2k},$$

where $(\mu^{2k})^*$ denotes the outer measure associated with the product measure μ^{2k} on X^{2k}.

The outer measure is needed in Definition 6.8 because $D_k(A, E, a, b)$ need not be measurable. The set $D_k(A, E, a, b)$ is measurable when A is a countable set of μ-measurable functions, since it is then a countable union of measurable sets.

Remarks 6.9. Let X, μ, $A \subseteq \mathbb{R}^X$ and $E \subseteq X$ be as above, and let $k \geq 1$ be an integer and $a < b$ be reals. The following hold.

(1) If $A \subseteq B \subseteq \mathbb{R}^X$, then
$$D_k(A, E, a, b) \subseteq D_k(B, E, a, b).$$
(2) If $a < a' < b' < b$, then
$$D_k(\overline{A}, E, a, b) \subseteq D_k(A, E, a', b').$$
[*Proof*: If $x \in D_k(\overline{A}, E, a, b)$, then there is $f \in \overline{A}$ such that $f(x_{2i}) \leq a < a'$ and $f(x_{2i+1}) \geq b > b'$ for all $i < k$. By the definition of the topology of pointwise convergence, there is $g \in A$ such that $g(x_{2i}) < a'$ and $g(x_{2i+1}) > b'$ for all $i < k$; hence $x \in D_k(A, E, a', b')$.]
(3) Every subset of a Talagrand μ-stable set is Talagrand μ-stable, by (1).
(4) If A is Talagrand μ-stable, then so is $\overline{A}$, by (2).

In Corollary 6.14 below we show that the deep computations of a confined set of computations are universally Monte Carlo computable when, for every feature and every type shard, the corresponding set of functions is Talagrand μ-stable for every Radon probability measure μ, that is, *universally Talagrand stable* in the sense of Definition 6.12 below. The proof of the corollary rests on Lemma 6.11, by which the pointwise closure of a Talagrand μ-stable set consists of μ-measurable functions. The lemma, in turn, rests on the following characterization of measurable functions (compare Fact 1.2).

Fact 6.10 ([Fre03, Lemma 413G])**.** *Let (X, Σ, μ) be a measure space, and let $\mathcal{K} \subseteq \Sigma$ be a collection of measurable sets such that the following conditions hold:*

- *(X, Σ, μ) is complete, i.e., for all $E \in \Sigma$ with $\mu(E) = 0$ and $F \subseteq E$ we have $F \in \Sigma$.*
- *(X, Σ, μ) is semi-finite, i.e., for all $E \in \Sigma$ with $\mu(E) = \infty$ there exists $F \subseteq E$ such that $F \in \Sigma$ and $0 < \mu(F) < \infty$.*
- *(X, Σ, μ) is saturated, i.e., $E \in \Sigma$ if and only if $E \cap F \in \Sigma$ for all $F \in \Sigma$ with $\mu(F) < \infty$.*

- (X, Σ, μ) *is inner regular with respect to* $\mathcal{K}$*, i.e., for all* $E \in \Sigma$

$$\mu(E) = \sup\{\mu(K) : K \in \mathcal{K} \text{ and } K \subseteq E\}.$$

(In particular, if X *is compact Hausdorff,* $\mathcal{K}$ *is the collection of compact subsets of* X*,* μ *is a Radon probability measure on* X*, and* Σ *is the completion of the Borel* σ*-algebra by* μ*, then all these conditions hold.) Then a function* $f\colon X \to \mathbb{R}$ *is measurable if and only if, for every* $K \in \mathcal{K}$ *with* $0 < \mu(K) < \infty$ *and all reals* $a < b$*, either* $\mu^*(K_{\leq a}) < \mu(K)$ *or* $\mu^*(K_{\geq b}) < \mu(K)$*, where* $K_{\leq a} = \{x \in K : f(x) \leq a\}$ *and* $K_{\geq b} = \{x \in K : f(x) \geq b\}$*.*

We first show that Talagrand stability passes to the pointwise closure and forces measurability.

Lemma 6.11. *Let* X *be a compact Hausdorff space, let* μ *be a Radon probability measure on* X*, and let* $A \subseteq \mathbb{R}^X$ *be a set of functions. If* A *is Talagrand* μ*-stable, then* $\overline{A}$ *is also Talagrand* μ*-stable and* $\overline{A} \subseteq \mathscr{M}^0(X, \mu)$*.*

Proof. By Remarks 6.9, the closure $\overline{A}$ is Talagrand μ-stable, and hence so is $\{f\}$ for every $f \in \overline{A}$. It therefore suffices to show that if $\{f\}$ is Talagrand μ-stable, then f is μ-measurable. Suppose that f is not μ-measurable. Then Fact 6.10 gives a compact set E of positive measure and reals $a < b$ with $\mu^*(E_{\leq a}) = \mu^*(E_{\geq b}) = \mu(E)$, where $E_{\leq a} = \{x \in E : f(x) \leq a\}$ and $E_{\geq b} = \{x \in E : f(x) \geq b\}$. For every $k \geq 1$ we have $(E_{\leq a} \times E_{\geq b})^k \subseteq D_k(\{f\}, E, a, b)$, and this product has outer measure $(\mu(E))^{2k}$ (see [Tal84]); hence $\{f\}$ is not Talagrand μ-stable. □

Definition 6.12. The set A is *universally Talagrand stable* if it is Talagrand μ-stable for every Radon probability measure μ on X.

For pointwise bounded sets of continuous functions on a compact space, universal Talagrand stability implies the NIP.

Proposition 6.13. *Let* X *be a compact Hausdorff space, and let* $A \subseteq \mathrm{C_p}(X)$ *be pointwise bounded. If* A *is universally Talagrand stable, then* A *satisfies the NIP.*

Proof. By Lemma 6.11, the closure $\overline{A}$ is contained in $\mathscr{M}^0(X, \mu)$ for every Radon probability measure μ on X, that is, $\overline{A} \subseteq M_r(X)$. By the implication (i)⇒(ii) of Theorem 6.5, applied with $K = X$, the set A satisfies the NIP. □

Corollary 6.14. *Let* $(L, \mathcal{P}, \Gamma)$ *be a CCS satisfying the Extendibility Axiom, let* R *be an exhaustive collection of sizers, and let* $\Delta \subseteq \Gamma$ *be* R*-confined. If* $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ *is universally Talagrand stable for all* $P \in \mathcal{P}$ *and all* $r_\bullet \in R$*, then every deep computation in* $\overline{\Delta}$ *is universally Monte Carlo computable on* R*.*

Proof. Let $P \in \mathcal{P}$ and $r_\bullet \in R$ be given. By the Extendibility Axiom and confinement, $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ is a pointwise bounded set of continuous functions on the compact space $\mathcal{L}[r_\bullet]$, and, by Proposition 6.13, it has the NIP; hence so does its restriction $\pi_P \circ \Delta|_{L[r_\bullet]}$. Theorem 6.7 now completes the proof. □

In the context of deep computations we have thus found two routes to universal Monte Carlo computability, the NIP and universal Talagrand stability, and the second implies the first (see Proposition 6.13). For countable sets of measurable functions, the following two results show that whether the NIP implies universal Talagrand stability depends on the axioms of set theory. On the one hand, it is consistent with the Zermelo–Fraenkel axioms of set theory with the axiom of choice (ZFC) that, for such sets, the NIP implies universal Talagrand stability.

Theorem 6.15 (Talagrand [Tal84, Theorem 9-3-1(a)])**.** *Let X be a compact Hausdorff space, and let $A \subseteq M_r(X)$ be countable and pointwise bounded. Assume that $[0,1]$ is not the union of fewer than $\mathfrak{c}$ closed sets of Lebesgue measure zero. If A satisfies the NIP, then A is universally Talagrand stable.*

(The hypothesis on $[0,1]$ follows, for example, from the Continuum Hypothesis.)

On the other hand, for Lebesgue measure, the failure of this implication is also consistent.

Theorem 6.16 (Fremlin and Shelah [FS93])**.** *It is consistent with ZFC that there exists a countable pointwise bounded set of Lebesgue measurable functions with the NIP that is not Talagrand stable with respect to Lebesgue measure.*

Both results concern countable sets of measurable functions. By the Extendibility Axiom, computations are continuous on the type shards, and continuous functions are measurable; hence, by Theorem 6.15, it is consistent that the NIP implies universal Talagrand stability for countable sets of computations. The example of Fremlin and Shelah, however, is a set of measurable functions, and it does not settle whether some countable set of computations can consistently have the NIP without being universally Talagrand stable; we leave this open (Question 7.4).

7. Open questions

We close with the questions that the paper leaves open, in the order in which they arise from its sections. The first concerns the hierarchy of Section 3.2, whose levels were defined through the topology of the space of deep computations; Proposition 3.4 describes them by the number of tests needed to tell deep computations apart, but these descriptions still involve infinitely many tests.

Question 7.1. Is there a finitary, combinatorial characterization of NIP_1, NIP_2 and NIP_3, in the spirit of the characterization of NIP by finite patterns in Theorem 3.1?

In Section 5 we place a single iterated map at every level of the hierarchy except one: at the level NIP but not NIP_1, both of our examples are actions of groups with two generators (see Section 5.5).

Question 7.2. Is there a cascade CCS that is NIP but not NIP_1? Equivalently, is there a continuous map γ of a compact metrizable space L such that (L,γ) is tame and $E(L,\gamma)$ is not first countable?

For actions of $\mathbb{Z}$, and more generally of abelian groups, the analogous question is Question 6.8 of Glasner and Megrelishvili [GM22].

Theorem 6.7 shows that NIP suffices for universal Monte Carlo computability; whether it is also necessary is the next question.

Question 7.3. Under the assumptions of Theorem 6.7, suppose that every deep computation in $\overline{\Delta}$ is universally Monte Carlo computable on R. Must $\pi_P \circ \Delta|_{L[r_\bullet]}$ have the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$?

Finally, in Section 6 we obtained two routes to universal Monte Carlo computability, the NIP and universal Talagrand stability; the second implies the first, and the results of Talagrand and of Fremlin and Shelah leave open whether, for computations, the first can hold without the second.

Question 7.4. Is it consistent with ZFC that some countable, pointwise bounded set of continuous functions on a compact Hausdorff space satisfies the NIP but is not universally Talagrand stable? Can this happen, consistently, for the sets $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ arising from a countable set Δ of computations as in Theorem 6.7?

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